Quasi-cyclic MDS classical and quantum codes from isogeny pullbacks of elliptic codes

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Abstract

Isogenies of elliptic curves are central to post-quantum cryptography, but have largely played no role in coding theory to date. We construct *pullback codes* by pulling back an elliptic code along a separable cyclic isogeny. Pullback codes are quasi-cyclic, preserve the rate and relative designed distance of the original code, and inherit iso-duality and self-orthogonality. When the degree divides $q-1$, a pullback code is equivalent to a matrix-product code built from twisted elliptic codes on the base curve. Using a subset-sum argument, we construct long quasi-cyclic MDS elliptic codes. We then show that the hull of a pullback code splits along the twists. Applying this to hull variation yields quasi-cyclic MDS entanglement-assisted quantum codes with prescribed entanglement.

Keywords: elliptic curves, isogenies, algebraic geometry codes, quasi-cyclic codes, matrix-product codes, MDS codes, hulls, entanglement-assisted quantum codes

1 Introduction

Algebraic geometry (AG) codes were introduced by Goppa [1] in the early 1980s. Their codewords are evaluations of functions in a Riemann–Roch space at rational points of a curve. They generalise the generalised Reed–Solomon (GRS) codes [2] and have Singleton defect at most the genus of the curve [3]. They garnered strong interest after some families of AG codes were shown to beat the Gilbert–Varshamov bound [4].

Elliptic curves give the simplest AG codes inequivalent to GRS codes. Elliptic curves are distinguished by the fact that their rational points carry a canonical abelian group structure. Consequently, conditions on divisors can often be translated into conditions on sums of points with respect to this group law. This reduces the MDS property to a subset-sum problem [5, 6], and has led to MDS [7], iso-dual [8] and self-dual [9] elliptic codes, with applications to quantum codes [8, 10].

Quasi-cyclic (QC) codes [11] arise on AG codes from curve automorphisms. Tong and Ding [12] built QC elliptic codes from automorphisms fixing the point at infinity, which have order at most 6. Although translations [13] can have larger orders, they do not fix the point at infinity, so the usual one- and two-point divisors are not preserved by these automorphisms.

An isogeny is a morphism between elliptic curves that preserves their group structures. In addition to playing a central role in post-quantum cryptography [14, 15], isogenies provide a natural way to construct divisors that are invariant under translations. Despite their cryptographic importance, however, they have hardly been used in coding theory. Given a separable isogeny $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ with cyclic kernel K of order ℓ and an elliptic code $C = C_{\mathcal{L}}(D, G)$ on $\mathcal{E}_2$, we study the **pullback code** $\varphi^*C := C_{\mathcal{L}}(\varphi^*D, \varphi^*G)$. Translations by K preserve φ^*D and φ^*G, and hence act on φ^*C. We show the following.

- φ^*C is an $[\ell n, \ell k, \geq \ell(n-k)]_q$ QC code of co-index ℓ (Theorem 11), whose K-fixed subcode is the ℓ-fold repetition of C (Proposition 13), and which inherits iso-duality from C (Corollary 12).
- If $\ell \mid q-1$, then φ^*C is monomially equivalent to a matrix-product code whose constituents are scalings of twists $C_{\mathcal{L}}(D, G_j)$ of C (Theorem 16). If φ^*C is MDS, so is every twist (Corollary 17).
- Modifying the index-two method of [7], pullbacks

give QC MDS elliptic codes (Construction 18, Example 19).

- The hull of φ^*C splits along its twists, and scaling whole fibres gives QC EAQECCs with prescribed entanglement (Theorems 22 and 26, Corollary 27).

Section 2 gives background, Section 3 collects tools from the group law, and Sections 4 and 5 contain our results.

2 Preliminaries

We summarise key definitions and results pertaining to divisors on algebraic curves, elliptic curves and their isogenies, and AG codes. For a more detailed description with proofs of results, we refer the reader to the following comprehensive standard references, which our exposition is based on:

- Silverman's book [16] for basic arithmetic geometry, with a focus on elliptic curves;
- Stichtenoth's book [3] for algebraic function fields and algebraic geometry (AG) codes.

Henceforth, let $q = p^m$ denote a prime power.

2.1 Divisors

Let $\mathcal{C}$ be a smooth projective, geometrically irreducible curve of genus g over $\mathbb{F}_q$, which we identify with its set of $\overline{\mathbb{F}}_q$-points. A point of $\mathcal{C}$ is **$\mathbb{F}_q$-rational** if its coordinates lie in $\mathbb{F}_q$, and the set of $\mathbb{F}_q$-rational points is denoted by $\mathcal{C}(\mathbb{F}_q)$.

A **divisor** of $\mathcal{C}$ is an element D of the free abelian group $\operatorname{Div}\mathcal{C} = \mathbb{Z}[\mathcal{C}]$ over the points of $\mathcal{C}$. In other words, it is a formal sum

$$D = \sum_{P\in\mathcal{C}} n_P \cdot (P), \quad n_P \in \mathbb{Z}$$

where all but finitely many n_P are zero. Its **support** and **degree** are, respectively,

$$\operatorname{supp} D = \{P \in \mathcal{C} : n_P \neq 0\} \quad \text{and} \quad \deg D = \sum_{P\in\mathcal{C}} n_P.$$

A partial ordering on $\operatorname{Div}\mathcal{C}$ is defined by

$$\sum_{P\in\mathcal{C}} n_P(P) \leq \sum_{P\in\mathcal{C}} n'_P(P) \iff \forall\, P \in \mathcal{C}: \quad n_P \leq n'_P.$$

D is called **effective** if $D \geq 0$.

The **function field** $\mathbb{F}_q(\mathcal{C})$ of $\mathcal{C}$ is the field of rational functions on $\mathcal{C}$ defined over $\mathbb{F}_q$. For $f \in \mathbb{F}_q(\mathcal{C})^*$, set

$$\operatorname{div} f := \sum_{P\in\mathcal{C}} \operatorname{ord}_P(f)\cdot(P), \quad \text{where}$$

- $\operatorname{ord}_P(f) = n \in \mathbb{Z}^+$ if f has a zero of order n at P;
- $\operatorname{ord}_P(f) = -n \in \mathbb{Z}^-$ if f has a pole of order n at P;
- $\operatorname{ord}_P(f) = 0$ otherwise.

Such divisors are called **principal** and form a subgroup $\operatorname{PDiv}\mathcal{C}$ of the degree-0 divisors $\operatorname{Div}^0\mathcal{C}$. Two divisors (of any degree) are called **equivalent**, written $A \sim B$, if they differ by a principal divisor. The **Jacobian** of $\mathcal{C}$ is the quotient $\operatorname{Jac}\mathcal{C} := \operatorname{Div}^0\mathcal{C}/\operatorname{PDiv}\mathcal{C}$.[1]

Given a divisor $G \in \operatorname{Div}\mathcal{C}$, we construct the **Riemann–Roch space**

$$\mathcal{L}(G) := \{f \in \mathbb{F}_q(\mathcal{C})^* : \operatorname{div} f \geq -G\} \cup \{0\}.$$

This is a finite-dimensional vector subspace of $\mathbb{F}_q(\mathcal{C})$, and we denote its dimension by $\ell(G)$. The Riemann–Roch Theorem [3, §1.5] gives us $\ell(G) \geq \deg G + 1 - g$, with equality when $\deg G > 2g - 2$.

The **space of differential forms** $\Omega_\mathcal{C}$ on $\mathcal{C}$ is the $\mathbb{F}_q$-vector space generated by symbols of the form $\mathrm{d}f$ for $f \in \mathbb{F}_q(\mathcal{C})$, subject to the relations

$$\mathrm{d}(f+g) = \mathrm{d}f + \mathrm{d}g, \quad \mathrm{d}(fg) = f\,\mathrm{d}g + g\,\mathrm{d}f, \quad \mathrm{d}a = 0$$

for all $f, g \in \mathbb{F}_q(\mathcal{C})$ and $a \in \mathbb{F}_q$. Around a point $P \in \mathcal{C}$, every $\omega \in \Omega^*_\mathcal{C}$ can be written uniquely as

$$\omega = \sum_{n=m}^{\infty} a_n t^n \mathrm{d}t \quad \text{for some} \begin{cases} t \in \mathbb{F}_q(\mathcal{C}),\ \operatorname{ord}_P(t) = 1, \\ a_n \in \mathbb{F}_q,\ a_m \neq 0,\ m \in \mathbb{Z}. \end{cases}$$

For $\omega \in \Omega^*_\mathcal{C}$, set $\operatorname{ord}_P(\omega) := m$, $\operatorname{Res}_P(\omega) := a_{-1}$ (or 0 if $m \geq 0$),

$$\operatorname{div}\omega := \sum_{P\in\mathcal{C}} \operatorname{ord}_P(\omega)\cdot(P), \quad \text{and}$$

$$\Omega(G) := \{\omega \in \Omega^*_\mathcal{C} : \operatorname{div}\omega \geq G\} \cup \{0\}.$$

Such divisors $\operatorname{div}\omega$ are called **canonical**. They all lie in one equivalence class and have degree $2g - 2$ [3, §1.5].

Finally, let $\varphi : \mathcal{C}_1 \to \mathcal{C}_2$ be a nonconstant morphism of curves, defined over $\mathbb{F}_q$. It induces a **pullback** map φ^* on functions, differentials, and divisors, defined by

$$\begin{aligned} \mathbb{F}_q(\mathcal{C}_2) &\to \mathbb{F}_q(\mathcal{C}_1), & f &\mapsto f \circ \varphi \\ \Omega_{\mathcal{C}_2} &\to \Omega_{\mathcal{C}_1}, & \mathrm{d}f &\mapsto \mathrm{d}(f \circ \varphi) \end{aligned}$$

[1] More precisely, this quotient is called the Picard group of degree 0, but this is isomorphic to the Jacobian variety.

$$\operatorname{Div}\mathcal{C}_2 \to \operatorname{Div}\mathcal{C}_1, \quad \sum_{Q\in\mathcal{C}_2} n_Q(Q) \mapsto \sum_{Q\in\mathcal{C}_2} n_Q \sum_{P\in\varphi^{-1}(Q)} e_P(P)$$

where e_P denotes the ramification index of $P \in \mathcal{C}_1$ in $\varphi^{-1}(Q)$, the **fibre** over $Q \in \mathcal{C}_2$. On functions, φ^* is an injective field homomorphism, so φ induces a finite field extension $\varphi^*\mathbb{F}_q(\mathcal{C}_2) \le \mathbb{F}_q(\mathcal{C}_1)$, whose degree is the **degree** $\deg\varphi$ of the morphism. We call φ **separable** if this extension is separable.

Some key properties of φ^*, which we use freely in our proofs, are as follows.

- On divisors, φ^* preserves the partial order and multiplies degrees by $\deg\varphi$.
- φ^* is compatible with principal divisors, in the sense that $\varphi^* \operatorname{div} f = \operatorname{div}\varphi^* f$.
- φ^* sends equivalent divisors to equivalent divisors.
- φ^* induces a homomorphism $\operatorname{Jac}\mathcal{C}_2 \to \operatorname{Jac}\mathcal{C}_1$.
- φ^* maps $\mathcal{L}(G)$ into $\mathcal{L}(\varphi^*G)$ for every $G \in \operatorname{Div}\mathcal{C}_2$.
- If φ is an automorphism of $\mathcal{C}$, then $\varphi^*(P) = \left(\varphi^{-1}(P)\right)$, so φ^* merely permutes the points of a divisor and preserves effectiveness.

2.2 Elliptic curves

An **elliptic curve** $\mathcal{E}$ over $\mathbb{F}_q$ is a smooth projective curve of genus 1 with an $\mathbb{F}_q$-rational point. Equivalently, $\mathcal{E}$ is the set of points $(x,y) \in \overline{\mathbb{F}}_q^2$ satisfying an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in \mathbb{F}_q \quad (1)$$

together with a point at infinity ∞, provided that no point is singular. Denote $\mathcal{E}(\mathbb{F}_q) := (\mathcal{E} \cap \mathbb{F}_q^2) \cup \{\infty\}$ and $\mathcal{E}(\mathbb{F}_q)^* := \mathcal{E} \cap \mathbb{F}_q^2$ for the set of all (resp. affine) $\mathbb{F}_q$-rational points. The cardinality $N := \#\mathcal{E}(\mathbb{F}_q)$ is constrained by the Hasse–Weil bound [16, § V.1]

$$|N - (q+1)| \le \lfloor 2\sqrt{q} \rfloor.$$

The points of $\mathcal{E}$ form an abelian group $(\mathcal{E}, \oplus)$ with identity ∞, where the group addition $\oplus$ is governed by the equations in [16, § III.2] and visualised in Figure 1. As is standard in the literature, $[m]P$ denotes the m-fold sum $P \oplus \cdots \oplus P$ for $m \in \mathbb{N}$, and $[-m]P := \ominus[m]P$.

We now specialise Section 2.1 to $\mathcal{C} = \mathcal{E}$, where $g = 1$. The function field $\mathbb{F}_q(\mathcal{E})$ is the field $\mathbb{F}_q(x,y)$, where x is transcendental over $\mathbb{F}_q$ and y is algebraic over

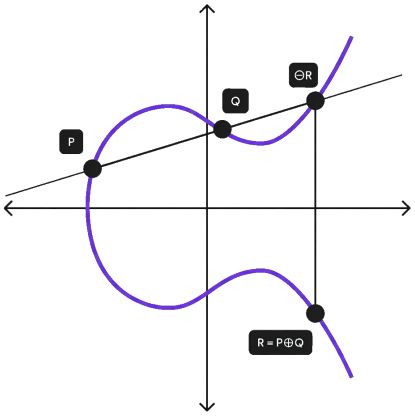


Figure 1. The group law of an elliptic curve $\mathcal{E}$ over $\mathbb{R}$ [17]

$\mathbb{F}_q(x)$ subject to Equation (1). The Riemann–Roch Theorem gives us $\ell(G) \ge \deg G$, with equality when $\deg G \ge 1$. The sets of canonical and principal divisors coincide [3, Cor. 1.4.12]. Moreover, the Jacobian $\operatorname{Jac}\mathcal{E}$ is famously isomorphic to $\mathcal{E}$ (cf. Section 3).

The **sum map** Σ on divisors evaluates the formal sum using the elliptic group law, and is denoted by

$$\Sigma(D) := \bigoplus_{P\in\mathcal{E}} [n_P]P.$$

2.3 Isogenies

Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be elliptic curves over $\mathbb{F}_q$. An **isogeny** $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ is a nonconstant morphism of curves that fixes ∞. Such a map is automatically a surjective group homomorphism with finite kernel. We assume henceforth that every isogeny is **separable**, so that $\ell = \#\ker\varphi$ and $e_P = 1$ for all P.

Every finite subgroup $K \le \mathcal{E}_1$ that is stable under the Frobenius map is the kernel of a separable isogeny $\mathcal{E}_1 \to \mathcal{E}_1/K$ defined over $\mathbb{F}_q$, which can be computed explicitly using Vélu's formulae [18]. In particular, every subgroup $K \le \mathcal{E}_1(\mathbb{F}_q)$ arises in this way.

Every isogeny φ admits a **dual isogeny** $\hat{\varphi} : \mathcal{E}_2 \to \mathcal{E}_1$, which satisfies $\hat{\varphi} \circ \varphi = [\ell]$ and $\varphi \circ \hat{\varphi} = [\ell]$. Its construction implies that $\Sigma(\varphi^*B) = \hat{\varphi}(\Sigma(B))$ for every $B \in \operatorname{Div}^0\mathcal{E}_2$.

For more details on isogenies and proofs of the above claims, see [16, § III.4–6].

2.4 Algebraic geometry codes

Let $n \in \mathbb{Z}^+$ and $D, G \in \operatorname{Div}\mathcal{C}$ be two $\mathbb{F}_q$-rational divisors (that is, divisors fixed by the q-power Frobenius)

such that $\operatorname{supp} D \cap \operatorname{supp} G = \varnothing$ and $D = (P_1)+\cdots+(P_n)$ is a sum of n distinct $\mathbb{F}_q$-rational points. Fixing an ordering of $\operatorname{supp} D$, we adopt the notation

$$f(D) := (f(P_1),\dots,f(P_n)) \in \mathbb{F}_q^n \quad \text{for} \quad f \in \mathcal{L}(G).$$

The **AG code** associated with D and G is

$$C_{\mathcal{L}}(D,G) := \{f(D) : f \in \mathcal{L}(G)\}.$$

If $\deg G < n$, then evaluation at D is injective on $\mathcal{L}(G)$, and $C_{\mathcal{L}}(D,G)$ is an $[n,k,d]_q$ code with $k = \ell(G) \geq \deg G + 1 - g$ and $d \geq n - \deg G$ [3, Thm. 2.2.2]. The quantity $d^* := n - \deg G$ is called the **designed distance**.

The Euclidean dual code $C_{\mathcal{L}}(D,G)^{\perp_E}$ of $C_{\mathcal{L}}(D,G)$ is

$$C_{\Omega}(D,G) = \left\{(\operatorname{Res}_{P_1}(\omega),\dots,\operatorname{Res}_{P_n}(\omega)) : \omega \in \Omega(G-D)\right\},$$

as proven in [19, Prop. 8]. Choose $t \in \mathbb{F}_q(\mathcal{C})$ such that $\operatorname{ord}_{P_i}(t) = 1$ for $1 \leq i \leq n$. Then, the differential $\eta := \mathrm{d}t/t$ has the property that every $\omega \in \Omega(G-D)$ can be written as $f\eta$ for some $f \in \mathcal{L}(H)$, where $H = D - G + \operatorname{div}\eta$ [3, §8.1]. As such, we have the beautiful result that $C_{\Omega}(D,G) = C_{\mathcal{L}}(D,H)$, so the dual of an AG code is again an AG code.

An **elliptic code** is an AG code on an elliptic curve $\mathcal{E}$ with $1 \leq \deg G \leq n-1$. By the above, it is an $[n,k,d]_q$ code with $k = \deg G$ and $d \geq n-k$, so its Singleton defect is at most 1. Its dual $C_{\mathcal{L}}(D,H)$ is also an elliptic code, as $\deg H = n-k$. A code is called **one-point** (resp. **two-point**) if $\operatorname{supp} G$ contains exactly one (resp. two) points.

We now recall some terminology for codes.

- For $\mathbf{v} \in (\mathbb{F}_q^*)^n$ and a code $C \subseteq \mathbb{F}_q^n$, write $\mathbf{v} \star C := \{(v_1c_1,\dots,v_nc_n) : \mathbf{c} \in C\}$ for the **scaling** of C by $\mathbf{v}$.
- Two codes are **monomially equivalent** if one is obtained from the other by a scaling and a coordinate permutation.
- An $[n,k,d]_q$ code is **maximum distance separable** (MDS) if $d = n-k+1$. It is **near-MDS** if $d = n-k$ and its dual also has Singleton defect 1 [20]. Hence, every elliptic code is either MDS or near-MDS.
- A code C is **self-dual** if $C^{\perp_E} = C$, and **iso-dual** if $C^{\perp_E} = \mathbf{v}\star C$ for some $\mathbf{v} \in (\mathbb{F}_q^*)^n$ [8, 9]. In both cases, we necessarily have $n = 2k$.
- A code of length mt is **quasi-cyclic** (QC) of index m and co-index t if it is invariant under a coordinate permutation consisting of m disjoint t-cycles [11, 12]. Equivalently, after reordering coordinates, it is invariant under m-fold cyclic shifts.
- Given codes $C_0,\dots,C_{s-1} \subseteq \mathbb{F}_q^n$ and a matrix $\mathbf{A} \in \mathbb{F}_q^{s\times t}$, the **matrix-product code** [21] is the code of length nt

$$(C_0,\dots,C_{s-1})\cdot\mathbf{A} := \left\{(\mathbf{c}_0,\dots,\mathbf{c}_{s-1})\mathbf{A} : \mathbf{c}_j \in C_j\right\},$$

where the $\mathbf{c}_j$ are column vectors, concatenated to form an $n \times s$ matrix.

2.5 Hulls and entanglement-assisted quantum codes

Throughout, $\perp$ denotes the Euclidean dual. The **hull** of a code C is $\operatorname{Hull}(C) := C \cap C^{\perp}$. The extreme cases $\operatorname{Hull}(C) = C$ and $\operatorname{Hull}(C) = \{\mathbf{0}\}$ are the **self-orthogonal** and **linear complementary dual** (LCD) codes respectively. The hull is not invariant under scaling, so equivalent codes may have hulls of different dimensions.

An **entanglement-assisted quantum error-correcting code** (EAQECC) with parameters $[\![n,\kappa,d;c]\!]_q$ encodes κ logical qudits into n physical qudits with the help of c pre-shared maximally entangled pairs, and has minimum distance d [22]. When $c = 0$, it is a quantum stabiliser code. The Singleton bound for EAQECCs reads

$$2d \leq n - \kappa + c + 2 \quad \text{whenever} \quad d \leq \tfrac{n}{2}+1,$$

and an EAQECC achieving it is called **MDS** [23]. EAQECCs can be built from any classical linear code, with the entanglement determined by the hull.

> **Lemma 2** (Hull bound [24, Lem. 16])
>
> Let C be an $[n,k,d]_q$ code and put $h = \dim \operatorname{Hull}(C)$. Then, there exists an EAQECC with parameters $[\![n,\ k-h,\ d;\ n-k-h]\!]_q$.
>
> If C is MDS and $d \leq \frac{n}{2}+1$ (equivalently, $k \geq \frac{n}{2}$), this EAQECC is also MDS.

3 Exploiting the group law

The group law on an elliptic curve allows conditions on divisors to be rewritten as conditions on sums of points. This section collects the consequences of this

principle that we will use as tools in Section 4. They are known, and we state them in the form we need.

Every curve $\mathcal{C}$ of genus g carries an abelian variety of dimension g, its Jacobian $\mathrm{Jac}\mathcal{C}$. The genus-one case is special because the group structure on the Jacobian can be transferred directly to the curve. Indeed, for an elliptic curve, the **Abel–Jacobi map**

$$\mathcal{E} \longrightarrow \mathrm{Jac}\,\mathcal{E}, \qquad P \longmapsto [(P)-(\infty)]$$

is a bijection, with inverse given by the sum map Σ of Section 2.2 The induced group law agrees with the usual geometric group law [16, § III.3]. In higher genera, the Jacobian is a higher-dimensional variety, and its group structure cannot be transferred to the curve in this way.

Lemma 3 (Abel–Jacobi [16, Cor. III.3.5])

$A \in \mathrm{Div}^0\,\mathcal{E}$ is principal if and only if $\Sigma(A) = \infty$.

As G is $\mathbb{F}_q$-rational, $\Sigma(G)$ is fixed by the Frobenius, so $\Sigma(G) \in \mathcal{E}(\mathbb{F}_q)$ even if the points of $\mathrm{supp}\,G$ are not themselves rational. The first tool decides whether an elliptic code is MDS or near-MDS. We use it to analyse MDS pullback codes in Corollary 17 and Construction 18.

Theorem 4 (Subset-sum criterion [5, 6], [25, Lem. 3.4])

Let $C_{\mathcal{L}}(D,G)$ be an elliptic code, with $D = (P_1)+\cdots+(P_n)$ and $k = \deg G$. Then $d = n-k$ if and only if there are k distinct points $P_{i_1},\dots,P_{i_k} \in \mathrm{supp}\,D$ with

$$P_{i_1} \oplus \cdots \oplus P_{i_k} = \Sigma(G). \tag{2}$$

Otherwise, $d = n-k+1$, and the code is MDS.

Indeed, if Equation (2) holds, then Lemma 3 gives a function $f \in \mathcal{L}(G)$ with $\mathrm{div}\,f = \sum_j (P_{i_j}) - G$, and $f(D)$ is a codeword of weight $n-k$.

The second tool lets us replace G by any divisor with the same degree and sum. We use it to identify the twists in Theorem 16 and to choose convenient divisors in Example 19.

Lemma 5 (Equivalent divisors [3, § 2.2])

If G and H are equivalent divisors whose supports are disjoint from D, then $C_{\mathcal{L}}(D,G)$ and $C_{\mathcal{L}}(D,H)$ differ at most by coordinate scaling.

Corollary 6 (Two-point normal form [7, 8])

Let $C_{\mathcal{L}}(D,G)$ be an elliptic code with $k = \deg G$. Let $Q_0, Q_1 \in \mathcal{E}(\mathbb{F}_q) \setminus \mathrm{supp}\,D$ be such that $G' := (k-1)(Q_0)+(Q_1)$ satisfies $\Sigma(G') = \Sigma(G)$. Then $C_{\mathcal{L}}(D,G)$ and $C_{\mathcal{L}}(D,G')$ differ at most by coordinate scaling.

Proof. $G - G'$ has degree 0 and sum ∞, so it is principal by Lemma 3. Apply Lemma 5. □

The third tool is a sufficient condition for iso-duality, which underlies the iso-dual MDS codes of [8]. We use it in Corollary 12.

Lemma 7 (Iso-dual criterion [8])

Let $C = C_{\mathcal{L}}(D,G)$ be an elliptic code. If $D - 2G$ is principal, or equivalently, if $\deg D = 2\deg G$ and $\Sigma(D) = [2]\Sigma(G)$, then C is iso-dual.

Proof. The two conditions are equivalent by Lemma 3. By Section 2.4, $C^{\perp_{\mathrm{E}}} = C_{\mathcal{L}}(D,H)$ with $H = D - G + \mathrm{div}\,\eta$. As $\mathrm{div}\,\eta$ is principal (Section 2.2), $H - G = D - 2G + \mathrm{div}\,\eta$ is principal, so $C_{\mathcal{L}}(D,H)$ is a scaling of C by Lemma 5. □

The fourth tool concerns automorphisms. For $R \in \mathcal{E}(\mathbb{F}_q)$, the **translation** map $\tau_R : P \mapsto P \oplus R$ is an automorphism of $\mathcal{E}$ as a curve. τ_R^* acts by

$$\mathbb{F}_q(\mathcal{E}) \ni f \mapsto f \circ \tau_R \quad \text{and} \quad \mathrm{Div}\,\mathcal{E} \ni (P) \mapsto (P \ominus R),$$

so that

$$\tau_R^*\mathcal{L}(G) = \mathcal{L}(\tau_R^* G) \ \text{ and } \ \Sigma(\tau_R^* G) = \Sigma(G) \ominus [\deg G]R.$$

Theorem 8 (Translation automorphisms [13])

Let $C := C_{\mathcal{L}}(D,G)$ be an elliptic code with $k = \deg G$. Suppose $R \in \mathcal{E}(\mathbb{F}_q)$ satisfies $[k]R = \infty$ and $\mathrm{supp}\,D$ is stable under τ_R.[a] Then τ_R induces a monomial automorphism of C.

[a] The latter condition holds, for instance, when $\mathrm{supp}\,D$ is a subgroup of $\mathcal{E}(\mathbb{F}_q)$ containing R, as in the index-two construction of [7].

Elliptic curves have many more curve automorphisms than curves of higher genus [13, 26], and these have been used to construct QC near-MDS codes [12] and optimal locally repairable codes [13]. Theorem 8 requires $[k]R = \infty$, as τ_R generally moves G to an inequivalent divisor. In Section 4, we avoid this restriction by choosing G to be invariant under the translations, which is exactly what pullbacks along isogenies achieve.

4 Pullback codes along isogenies

Isogenies have found important applications in cryptography, notably in the post-quantum signature scheme SQIsign [14] and related protocols [15, 27]. Their potential applications in coding theory, however, remain largely unexplored. In this section, we use isogenies to transform elliptic codes on one curve into longer elliptic codes on another, and study the properties of the resulting codes.

Given an isogeny $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ and an elliptic code $C_{\mathcal{L}}(D, G)$ on $\mathcal{E}_2$, the most natural code on $\mathcal{E}_1$ is obtained by pulling back both divisors. We impose mild constraints on φ so that the resulting code is nice.

Definition 9 (Pullback codes)

Let $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ be a separable isogeny of degree ℓ with cyclic kernel $K := \ker \varphi \subseteq \mathcal{E}_1(\mathbb{F}_q)$. Let $C := C_{\mathcal{L}}(D, G)$ be an $[n, k]_q$ elliptic code on $\mathcal{E}_2$ with

$$D = \textstyle\sum_{i=1}^{n} (P_i) \in \operatorname{Div} \mathcal{E}_2, \quad \varphi^{-1}(P_i) \subseteq \mathcal{E}_1(\mathbb{F}_q).$$

We call $\varphi^* C := C_{\mathcal{L}}(\varphi^* D, \varphi^* G)$ the **pullback code** of C along φ.

All results in the sequel assume the conditions and notation of Definition 9. We first record three facts from arithmetic geometry that subsequent results rely on.

Lemma 10 (Pullback facts)

(a) $\tau_T^* \varphi^* = \varphi^*$ for every $T \in K$.

(b) $\mathbb{F}_q(\mathcal{E}_1)/\varphi^* \mathbb{F}_q(\mathcal{E}_2)$ is Galois with group $\{\tau_T^* : T \in K\}$, so the functions fixed by every τ_T^* are exactly the pullbacks.

(c) For $A, B \in \operatorname{Div} \mathcal{E}_2$, we have $\varphi^* A \geq \varphi^* B$ if and only if $A \geq B$. A divisor on $\mathcal{E}_1$ whose coefficients are constant on cosets of K is a pullback.

Proof. (a) $\varphi(P \oplus T) = \varphi(P)$ for $T \in K$. (b) Follows from [16, Thm. III.4.10]. (c) As φ is separable, $\varphi^*(Q)$ is the sum of the ℓ distinct points of the fibre over Q. Distinct points have disjoint fibres. Comparing coefficients gives the claim. □

4.1 Basic properties

We begin with the parameters and quasi-cyclicity of $\varphi^* C$.

Theorem 11 (Quasi-cyclic pullback codes)

$\varphi^* C$ is an $[\ell n, \ell k, \geq \ell(n-k)]_q$ elliptic code on $\mathcal{E}_1$. Furthermore, $\varphi^* C$ is quasi-cyclic of co-index ℓ.

Proof. The fibres over the points of D form n disjoint cosets of K in $\mathcal{E}_1(\mathbb{F}_q)$, each with size ℓ. Thus, $\varphi^* D$ is a sum of ℓn distinct rational points, disjoint from $\operatorname{supp} \varphi^* G$. As $\deg \varphi^* G = \ell k$, the code $\varphi^* C$ is elliptic with the stated parameters by Section 2.4.

Now, we show that $\varphi^* C$ is quasi-cyclic of co-index ℓ. Let $K = \langle T \rangle$. By [3, Prop. 8.2.3(a)], an automorphism σ of $\mathcal{E}_1$ that fixes both $\varphi^* D$ and $\varphi^* G$ acts on $\varphi^* C$ by permuting coordinates, via $f(Q) \mapsto f(\sigma(Q))$ for $f \in \mathcal{L}(\varphi^* G)$. The translation τ_T fixes $\varphi^* G$ and $\varphi^* D$ by Lemma 10(a), so it permutes $\operatorname{supp} \varphi^* D$ by cycling the points of each fibre. As $\operatorname{ord} T = \ell$, this permutation consists of n disjoint ℓ-cycles, so $\varphi^* C$ is quasi-cyclic of co-index ℓ. □

Next, we show that $\varphi^* C$ inherits iso-duality from C.

Corollary 12 (Iso-dual pullback codes)

If $D - 2G$ is principal, then C and $\varphi^* C$ are both iso-dual.

Proof. Since $\varphi^* \operatorname{div} f = \operatorname{div} \varphi^* f$, if $D - 2G$ is principal, then so is $\varphi^* D - 2\varphi^* G$. Hence, Lemma 7 applies to both C and $\varphi^* C$. □

The translations by K act on $\varphi^* C$, and the codewords they fix come from C itself. Let

$$C_{\varphi^*\mathcal{L}}(\varphi^*D, G) := \{(\varphi^*f)(\varphi^*D) : f \in \mathcal{L}(G)\}.$$

As $(\varphi^*f)(Q) = f(\varphi(Q))$, this code is the ℓ-fold repetition of C, in which each coordinate is repeated along its fibre.

Proposition 13 (Fixed subcode)

The subcode of φ^*C fixed by K is $C_{\varphi^*\mathcal{L}}(\varphi^*D, G)$, which has parameters $[\ell n, k, \ell d]$, where d is the minimum distance of C.

Proof. Let $T \in K$. First, we prove the backward inclusion. By Section 2.1, $\varphi^*\mathcal{L}(G) \subseteq \mathcal{L}(\varphi^*G)$, and each $\varphi^*f = f \circ \varphi = f \circ \varphi \circ \tau_T$ is constant on fibres. So, $C_{\varphi^*\mathcal{L}}(\varphi^*D, G) \subseteq \varphi^*C$ is fixed by K.

Now, we prove the forward inclusion. Suppose $\mathbf{c} := f(\varphi^*D) \in \varphi^*C$ is fixed by K. In other words, the function $\tau_T^*f - f \in \mathcal{L}(\varphi^*G)$ vanishes on φ^*D, which has $\ell n > \deg \varphi^*G$ points. Hence, $\tau_T^*f - f$ must be identically zero, i.e. f is fixed by τ_T^*. By Lemma 10(b), $f \in \varphi^*\mathbb{F}_q(\mathcal{E}_2)$, i.e. $f = \varphi^*g$ for some $g \in \mathbb{F}_q(\mathcal{E}_2)$. But

$$f = \varphi^*g \in \mathcal{L}(\varphi^*G) \implies \operatorname{div}\varphi^*g \geq -\varphi^*G$$
$$\implies \operatorname{div} g \geq -G \implies g \in \mathcal{L}(G),$$

the second implication by Lemma 10(c). Hence, $\mathbf{c} = \varphi^*g(\varphi^*D) \in C_{\varphi^*\mathcal{L}}(\varphi^*D, G)$. □

For our construction to be useful in practice, a basis of $\mathcal{L}(\varphi^*G)$ must be easy to compute. Fortunately, this is the case for the usual one-point divisor $G = k(\infty)$, as Proposition 14 shows. The basis only needs the x-coordinates of the kernel points, and not the explicit isogeny φ.

Proposition 14 (Basis of $\mathcal{L}(\varphi^*G)$)

Suppose ℓ is odd and $G = k(\infty)$. Label the $\ell - 1$ points of $K \setminus \{\infty\}$ by $\oplus T_1, \ldots, \oplus T_{(\ell-1)/2}$, and put $z := (x - x(T_1)) \cdots (x - x(T_{(\ell-1)/2}))$. Then, $\mathcal{L}(\varphi^*G) = z^{-k}\mathcal{L}(\ell k(\infty))$, which has basis

$$\left\{x^i y^j z^{-k} : j \in \{0,1\},\ 2i + 3j \leq \ell k\right\}.$$

Proof. As ℓ is odd, each T_r has order > 2, so the $x(T_r)$ are distinct. From $\operatorname{div}(x - x(T_r)) = (T_r) + (\ominus T_r) - 2(\infty)$ and $\varphi^*G = k\sum_{T' \in K}(T')$, we obtain

$$\operatorname{div}(z^k) = k \sum_{T' \in K \setminus \{\infty\}} (T') - k(\ell - 1)(\infty) = \varphi^*G - \ell k(\infty).$$

Adding $\operatorname{div} f + \ell k(\infty)$ to both sides gives

$$\operatorname{div}(z^k f) + \ell k(\infty) = \operatorname{div} f + \varphi^*G \quad \forall f \in \mathbb{F}_q(\mathcal{E}_1).$$

Hence, $f \in \mathcal{L}(\varphi^*G)$ if and only if $z^k f \in \mathcal{L}(\ell k(\infty))$. The functions $x^i y^j$ with $j \in \{0,1\}$ and $2i + 3j \leq \ell k$ form a basis of $\mathcal{L}(\ell k(\infty))$ [16, §III.3], since x and y have poles of order 2 and 3 respectively at ∞. Dividing by z^k gives the claimed basis. □

As $\operatorname{supp}\varphi^*D$ is disjoint from K, the function z has no zeros on $\operatorname{supp}\varphi^*D$. So, evaluating this basis at φ^*D gives a generator matrix of φ^*C directly. The main computational cost of φ^*C thus lies in finding the fibres $\varphi^{-1}(P_i)$, not in the Riemann–Roch space.

4.2 Splitting into twists

The translation $\tau := \tau_T^*$ is a linear operator of order ℓ on $\mathcal{L}(\varphi^*G)$. When $\ell \mid q - 1$, it is diagonalisable over $\mathbb{F}_q$, and its eigenspaces decompose φ^*C into codes of length n on $\mathcal{E}_2$. We first describe the eigenspaces, using tools from arithmetic geometry.

Lemma 15 (Eigenfunctions of translations)

In Definition 9, suppose $\ell \mid q - 1$ and let $\zeta \in \mathbb{F}_q$ have order ℓ. Write $K = \langle T \rangle$ and $\tau := \tau_T^*$. Then, there exist an eigenfunction $u \in \mathbb{F}_q(\mathcal{E}_1)^*$ and a divisor $A \in \operatorname{Div}^0 \mathcal{E}_2$ with

(a) $\tau u = \zeta u$, $\operatorname{div} u = \varphi^*A$, and $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$;

(b) $R := \Sigma(A)$ generates $\ker\hat{\varphi}$, which is therefore rational;

(c) for every $j \in \mathbb{Z}$, the ζ^j-eigenspace of τ on $\mathcal{L}(\varphi^*G)$ is $u^j\varphi^*\mathcal{L}(G + jA)$.

Proof. (a) By Lemma 10(b), $\mathbb{F}_q(\mathcal{E}_1)/\varphi^*\mathbb{F}_q(\mathcal{E}_2)$ is cyclic of degree ℓ, generated by τ. As ζ^{-1} has norm $\zeta^{-\ell} = 1$, Hilbert's Theorem 90 [28, Thm. VI.6.1] gives $u \in \mathbb{F}_q(\mathcal{E}_1)^*$ with $\zeta^{-1} = u/\tau(u)$, i.e. $\tau(u) = \zeta u$. Since

$$\tau(\operatorname{div} u) = \operatorname{div}(\tau(u)) = \operatorname{div}(\zeta u) = \operatorname{div} u,$$

$\operatorname{div} u$ is fixed by K. Lemma 10(c) gives $\operatorname{div} u = \varphi^*A$ for some $A \in \operatorname{Div}\mathcal{E}_2$. Also, u^ℓ is fixed by τ, so $u^\ell = \varphi^*h$ for some $h \in \mathbb{F}_q(\mathcal{E}_2)$. Taking div on both sides gives

$$\operatorname{div}(u^\ell) = \ell \operatorname{div} u = \ell\varphi^*A = \varphi^*(\ell A) = \varphi^* \operatorname{div} h,$$

which forces $\ell A = \operatorname{div} h$. In particular, $\deg A = 0$.

By the approximation theorem [3, Thm. 1.3.1], replacing u by $u \cdot \varphi^* a$ for suitable $a \in \mathbb{F}_q(\mathcal{E}_2)^*$, we may assume $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$.

(b) Put $R := \Sigma(A) \in \mathcal{E}_2(\mathbb{F}_q)$. As $\varphi^* A = \operatorname{div} u$ is principal and $\deg A = 0$, we have $\hat{\varphi}(\Sigma(A)) = \Sigma(\varphi^* A) = \infty$ by Lemma 3, so $R \in \ker \hat{\varphi}$.

We show that R has order ℓ. Suppose $[m]R = \infty$ with $1 \le m < \ell$. Then, $mA = \operatorname{div} a$ for some a. Consider the function $v := u^m/\varphi^* a$. Its divisor is $\operatorname{div} v = m\varphi^*(A) - \varphi^*(mA) = 0$, so v is a nonzero constant, hence fixed by τ. But $\tau(v) = (\zeta u)^m/\varphi^* a = \zeta^m v$, which forces $\zeta^m = 1$, a contradiction. Thus, R has order ℓ.

As $\hat{\varphi} \circ \varphi = [\ell]$ is separable for $\ell \nmid q$, so is $\hat{\varphi}$, and $\#\ker \hat{\varphi} = \ell$. Hence R generates $\ker \hat{\varphi}$, which is therefore rational.

(c) By Lemma 10(b), a function $f \in \mathbb{F}_q(\mathcal{E}_1)$ satisfies $\tau f = \zeta^j f$ if and only if f/u^j is fixed by τ, that is, $f = u^j \varphi^* g$ for some $g \in \mathbb{F}_q(\mathcal{E}_2)$. For such f, we have $\operatorname{div} f = \varphi^*(jA + \operatorname{div} g)$, so by Lemma 10(c), $f \in \mathcal{L}(\varphi^* G)$ if and only if $jA + \operatorname{div} g \ge -G$, that is, $g \in \mathcal{L}(G + jA)$. □

We are now ready to express $\varphi^* C$ as a matrix-product code [21]. Each constituent is an elliptic code on $\mathcal{E}_2$ obtained from C by 'twisting' its defining divisor G but retaining the evaluation points D. We refer to these constituent codes as the **twists** of C.

Theorem 16 (Splitting into twists)

Adopt the notation and hypotheses of Lemma 15. Then, up to a reordering of $\varphi^* D$, $\varphi^* C$ is the matrix product code $(C_0, \dots, C_{\ell-1}) \cdot \mathbf{V}$ with $\mathbf{V} := [\zeta^{jm}]_{j,m=0}^{\ell-1}$, where each C_j is a scaling of the twist $C_{\mathcal{L}}(D, G + jA)$.

Up to scaling, one may replace $G + jA$ by any divisor G_j disjoint from D with $\deg G_j = k$ and $\Sigma(G_j) = \Sigma(G) \oplus [j]R$, where R is a suitable generator for $\langle R \rangle = \ker \hat{\varphi}$.

Proof. Write $\tau := \tau_T^*$, and fix $\zeta \in \mathbb{F}_q$ of order ℓ. As τ has order $\ell \nmid q$, the operator τ on $\mathcal{L}(\varphi^* G)$ has minimal polynomial dividing $X^\ell - 1$. This polynomial has roots ζ^j, $0 \le j \le \ell - 1$, so τ is diagonalisable with eigenvalues among the ζ^j. Henceforth, j denotes such an exponent. Let L_j be the ζ^j-eigenspace and W_j its image in $\varphi^* C$. Since evaluation at $\varphi^* D$ is injective, we have $\varphi^* C = \bigoplus_j W_j$.

Let u, A and R be as in Lemma 15, so $L_j = u^j \varphi^* \mathcal{L}(G + jA)$. Fix a choice of $\tilde{P}_i \in \varphi^{-1}(P_i)$ for each i, so that

$$\varphi^{-1}(P_i) = \left\{ \tilde{P}_i \oplus [m]T : m \in \mathbb{Z}/\ell\mathbb{Z} \right\}.$$

Order the coordinates of $\varphi^* C$ (and each W_j) as

$$\tilde{P}_1, \dots, \tilde{P}_n;\ \tilde{P}_1 \oplus T, \dots, \tilde{P}_n \oplus T;\ \tilde{P}_1 \oplus [2]T, \dots, \tilde{P}_n \oplus [2]T;\ \dots;\ \tilde{P}_1 \ominus T, \dots, \tilde{P}_n \ominus T.$$

Any $f \in L_j$ can be expressed as $f = u^j \varphi^* g$ for some $g \in \mathcal{L}(G + jA)$. We have $f(\tilde{P}_i) = u(\tilde{P}_i)^j g(P_i)$, where u has neither a zero nor a pole at $\tilde{P}_i$ as $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$. Hence, the restrictions of W_j to the coordinates $\tilde{P}_1, \dots, \tilde{P}_n$ form

$$C_j := \mathbf{v}_j \star C_{\mathcal{L}}(D, G + jA) \quad \text{where} \quad \mathbf{v}_j = \left(u(\tilde{P}_i)^j \right)_{i=1}^n.$$

Moreover,

$$f(\tilde{P}_i \oplus [m]T) = (\tau^m f)(\tilde{P}_i) = \zeta^{jm} f(\tilde{P}_i), \quad \text{so}$$
$$W_j = \left\{ \left(\mathbf{c}, \zeta^j \mathbf{c}, \cdots, \zeta^{j(\ell-1)} \mathbf{c} \right) : \mathbf{c} \in C_j \right\}.$$

Hence, $\varphi^* C = \bigoplus_j W_j = (C_0, \dots, C_{\ell-1}) \cdot \mathbf{V}$.

For the second part of the theorem statement, observe that $G + jA$ satisfies the same conditions as G_j, so they are equivalent by Lemma 3. Lemma 5 then asserts that $C_{\mathcal{L}}(D, G_j)$ is a scaling of $C_{\mathcal{L}}(D, G + jA)$. □

Next, we present a necessary condition for $\varphi^* C$ to be MDS. Unlike Theorem 4, which concerns sums of ℓk points of $\mathcal{E}_1$, Corollary 17 concerns ℓ sums of only k points of $\mathcal{E}_2$.

Corollary 17 (MDS twists)

If $\varphi^* C$ is MDS, then no k points of $\operatorname{supp} D$ sum into $\Sigma(G) \oplus \ker \hat{\varphi}$. In the setting of Theorem 16, every twist $C_{\mathcal{L}}(D, G + jA)$ is then MDS.

Proof. By Section 2.3, every divisor $B \in \operatorname{Div} \mathcal{E}_2$ of degree 0 satisfies $\Sigma(\varphi^* B) = \hat{\varphi}(\Sigma(B))$. Replacing B by $B - b(\infty)$, every divisor B of degree b satisfies

$$\Sigma(\varphi^* B) = \hat{\varphi}(\Sigma(B)) \oplus [b] s_K \quad \text{where} \quad s_K := \Sigma(\varphi^*(\infty)).$$

Suppose that distinct points $P_{i_1}, \dots, P_{i_k} \in \operatorname{supp} D$ satisfy $P_{i_1} \oplus \cdots \oplus P_{i_k} \in \Sigma(G) \oplus \ker \hat{\varphi}$. Applying the identity to both $B = \sum_j (P_{i_j})$ and G, which both have degree k,

we obtain

$$\Sigma\Big(\varphi^*\textstyle\sum_j(P_{i_j})\Big) = \hat{\varphi}\Big(P_{i_1}\oplus\cdots\oplus P_{i_k}\Big)\oplus[k]s_K$$
$$= \hat{\varphi}(\Sigma(G))\oplus[k]s_K = \Sigma(\varphi^*G).$$

Since $\varphi^*\sum_j(P_{i_j})$ contains ℓk distinct points of $\operatorname{supp}\varphi^*D$, φ^*C is not MDS by Theorem 4. The main result follows by contraposition.

In the setting of Theorem 16, we have $\ker\hat{\varphi} = \langle R\rangle$, so $\Sigma(G+jA) = \Sigma(G)\oplus[j]R$ runs over $\Sigma(G)\oplus\ker\hat{\varphi}$. Hence, if φ^*C is MDS, no k points of $\operatorname{supp}D$ sum to any $\Sigma(G+jA)$, and each twist $C_{\mathcal{L}}(D, G+jA)$ is MDS by Theorem 4. □

4.3 Comparison with previous work

To the best of our knowledge, isogenies have not previously been used to construct elliptic codes. The closest related work is that of Tong and Ding [12], who constructed quasi-cyclic elliptic codes using automorphisms of the underlying elliptic curve. Although both approaches produce quasi-cyclic codes, their underlying symmetries and resulting code properties differ substantially. We summarise these differences below in the format: [12] *vs* ours.

- **Symmetry:** automorphisms fixing ∞ *vs* translations by $\ker\varphi$
- **Co-index:** ≤ 6 *vs* any ℓ dividing the exponent of $\mathcal{E}_1(\mathbb{F}_q)$
- G**:** $k(\infty)$ *vs* a pullback divisor
- **Orbit sizes:** sometimes varied *vs* uniform (size ℓ)
- **MDS:** exclusively near-MDS *vs* possibly MDS

4.4 Applications

The results in the previous subsections have three immediate applications. The first is **longer iso-dual elliptic codes**. Zhu and Zhao [8, Constr. 1] construct iso-dual MDS codes of length $2k$ with

$$D = \sum_{i=1}^{k}\Big((Q\oplus P_i)+(Q\ominus P_i)\Big), \quad G = (k-1)(\infty)+(Q),$$

where Q has order 2. These codes satisfy the hypothesis of Corollary 12, since $\Sigma(D) = \infty = [2]Q = \Sigma(2G)$. If $\operatorname{supp}D\subseteq\varphi(\mathcal{E}_1(\mathbb{F}_q))$, then all fibres above D consist of rational points. The pullback construction consequently extends the original iso-dual MDS code of length $2k$ to an iso-dual code of length $2\ell k$, without changing its rate or relative designed distance. The resulting code is not necessarily MDS, however, and may incur a loss of one unit in minimum distance.

The second is **decoding via twists**. Ling and Solé showed that every quasi-cyclic code decomposes into shorter constituent codes [11]. Theorem 16 identifies the constituents of φ^*C as scalings of the twists $C_{\mathcal{L}}(D, G_j)$, which are $[n, k, \geq n-k]$ elliptic codes on $\mathcal{E}_2$. Each of the ℓ twists can be decoded independently up to half its minimum distance. Consequently, this approach corrects any error pattern supported on at most $\lfloor(n-k-1)/2\rfloor$ fibres $\varphi^{-1}(P_i)$, regardless of the number of errors within an individual fibre. In this setting, decoding reduces to ℓ instances of length n, instead of a single instance of length ℓn.

The third is a supply of **quasi-cyclic elliptic codes**. By Theorem 11, the pullback produces quasi-cyclic elliptic codes of co-index ℓ with the same rate $\frac{k}{n}$ and relative designed distance $\frac{n-k}{n}$ as C, provided $\mathcal{E}_1(\mathbb{F}_q)$ has a cyclic subgroup of order ℓ. We can combine this with the index-two construction of Wang et al. [7] to obtain **MDS quasi-cyclic elliptic codes**, provided that the index-two subgroup is compatible with the fibres of φ. In [7], D is supported on a subgroup $H\leq\mathcal{E}(\mathbb{F}_q)$ of index 2 or its non-trivial coset, and $\Sigma(G)$ is chosen in the other coset, so that Equation (2) fails by a parity argument. This is illustrated in Construction 18 and Example 19.

Construction 18 (MDS QC pullback codes)

Let $\varphi:\mathcal{E}_1\to\mathcal{E}_2$ and K be as in Definition 9, with ℓ odd. Let $N = \#\mathcal{E}_1(\mathbb{F}_q)$ be even, let $H\leq\mathcal{E}_1(\mathbb{F}_q)$ be a subgroup of index 2, and choose $Q\in\mathcal{E}_1(\mathbb{F}_q)\setminus H$.

(a) If k is odd with $\ell k < N/2$, set $D = \sum_{P\in\varphi(Q\oplus H)}(P)$ and $G = k(\infty)$. Then $\varphi^*C = C_{\mathcal{L}}(\varphi^*D, \varphi^*G)$ is an $[N/2,\ \ell k,\ N/2-\ell k+1]_q$ MDS code.

(b) For any $k\geq 1$ with $\ell k < N/2-\ell$, set $D = \sum_{P\in\varphi(H)\setminus\{\infty\}}(P)$ and $G = (k-1)(\infty)+(\varphi(Q))$. Then φ^*C is an $[N/2-\ell,\ \ell k,\ N/2-\ell-\ell k+1]_q$ MDS code.

In both cases, φ^*C is quasi-cyclic of co-index

ℓ.

Proof. We check that the codes C and φ^*C are well-defined, i.e. the supports of D and G (resp. φ^*D and φ^*G) are disjoint and $\mathbb{F}_q$-rational.

The image of K in $\mathcal{E}_1(\mathbb{F}_q)/H \cong \mathbb{Z}/2\mathbb{Z}$ has order dividing both $\#K = \ell$ and 2, so it is trivial, and $K \subseteq H$. Hence, H and $Q \oplus H$ are unions of cosets of K, i.e. they are partitioned by the fibres of φ. So $\varphi^{-1}(\varphi(H)) = H$, $\varphi^{-1}(\varphi(Q \oplus H)) = Q \oplus H$, and $\varphi(H) \cap \varphi(Q \oplus H) = \varnothing$. In (a), this shows that

$$\operatorname{supp} \varphi^*G = \varphi^{-1}(\infty) = K \subseteq \mathcal{E}_1(\mathbb{F}_q) \quad \text{and}$$
$$\operatorname{supp} \varphi^*D = \varphi^{-1}(\varphi(Q \oplus H)) \subseteq Q \oplus H \subseteq \mathcal{E}_1(\mathbb{F}_q)$$

are $\mathbb{F}_q$-rational. Likewise, in (b), both

$$\operatorname{supp} \varphi^*G = K \cup (Q \oplus K) \subseteq \mathcal{E}_1(\mathbb{F}_q) \quad \text{and}$$
$$\operatorname{supp} \varphi^*D = \varphi^{-1}(\varphi(H \setminus \{\infty\})) \subseteq H \subseteq \mathcal{E}_1(\mathbb{F}_q)$$

are $\mathbb{F}_q$-rational. As $\infty = \varphi(\infty) \in \varphi(H)$ and $\varphi(Q) \in \varphi(Q \oplus H)$, the supports of D and G are disjoint in both cases, so C is well-defined. It follows that the supports of φ^*D and φ^*G are also disjoint, so φ^*C is well-defined.

Now, we show that φ^*C is MDS by means of Theorem 4. Write $K = \langle T \rangle$ and $s_K := \sum_{T' \in K} T' = [\ell(\ell-1)/2]T$. Since ℓ is odd, $s_K = \infty$.

For (a), $\Sigma(\varphi^*G) = [k]s_K = \infty \in H$. But $\operatorname{supp} \varphi^*D = Q \oplus H$, so any ℓk of its points sum into $[\ell k]Q \oplus H = Q \oplus H$, as ℓk is odd.

For (b), the fibre over $\varphi(Q)$ is $Q \oplus K$, so $\Sigma(\varphi^*G) = [k-1]s_K \oplus [\ell]Q \oplus s_K = [\ell]Q$, which lies in $Q \oplus H$ as ℓ is odd. But $\operatorname{supp} \varphi^*D \subseteq H$, so any ℓk of its points sum into H.

In both cases, no ℓk points of $\operatorname{supp} \varphi^*D$ sum to $\Sigma(\varphi^*G)$, so φ^*C is MDS by Theorem 4. □

In Construction 18, one may replace D by a subdivisor since removing points from D preserves both arguments. The argument also extends to a subgroup H of any prime index $m \nmid \ell$, giving MDS codes of length N/m whenever $m \nmid \ell k$. This also covers even ℓ, for example $\ell = 4$ with $m = 3$. We conclude this section with an extended example.

Example 19 ($[144, 3k]_{256}$ MDS QC codes [29])

Let $\mathbb{F}_{256} = \mathbb{F}_2(\alpha)$ with $\alpha^8 = \alpha^4 + \alpha^3 + \alpha^2 + 1$. Consider $\mathcal{E}_1 : y^2 + xy = x^3 + 1$ over $\mathbb{F}_{256}$. Then, $N = \#\mathcal{E}_1(\mathbb{F}_{256}) = 288$. This is 1 off from the Hasse–Weil bound $M := q + 1 + \lfloor 2\sqrt{q} \rfloor = 289$. The point $(0, 1)$ is the only rational point of order 2, so $H := [2]\mathcal{E}_1(\mathbb{F}_{256})$ has index 2.

The point $T = (\alpha^{238}, \alpha^{176})$ has order $\ell = 3$. Vélu's formulae give the isogeny $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ with kernel $K = \langle T \rangle$, where

$$\mathcal{E}_2 : y^2 + xy = x^3 + \alpha^{238}x + \alpha^{51}.$$

As $3 \mid 255 = q - 1$, we may take $\zeta = \alpha^{85}$, an element of order 3 in $\mathbb{F}_{256}$. Then

$$u = \frac{y + \alpha^{123}x + \alpha^{34}}{x + \alpha^{238}}$$

is an eigenfunction of $\tau := \tau_T^*$ as in Lemma 15, with $\tau u = \zeta u$. We have $\operatorname{div} u = \varphi^*A$ with $A = (R) - (\infty)$, where $R := (\alpha^{51}, \alpha^{20})$. Here, $\mathcal{E}_1[3] \subseteq \mathcal{E}_1(\mathbb{F}_{256})$, and $\ker \hat{\varphi} = \langle R \rangle = \varphi(\mathcal{E}_1[3])$.

Let $Q \in \mathcal{E}_1(\mathbb{F}_{256}) \setminus H$, and let D be the sum of the 48 points of $\varphi(Q \oplus H)$. These are the points of $\mathcal{E}_2$ with $x = \alpha^e$ for

$$e \in \{5, 11, 57, 63, 79, 80, 88, 94, 109, 120, 133, 135, 147, 154, 169, 176, 188, 190, 203, 214, 229, 235, 243, 244\}.$$

Let $G = k(\infty)$ with k odd. By Construction 18(a), φ^*C is a quasi-cyclic $[144, 3k, 145 - 3k]_{256}$ MDS code of co-index 3. Its length is $\lfloor M/2 \rfloor = 144$. This is close to the upper bound $(N + 6)/2 = 147$ of Han and Ren [25, Thm. 1.2] on the length of any non-trivial MDS elliptic code on $\mathcal{E}_1$.

Next, we express φ^*C as a matrix-product code. Take $\tilde{P}_i \in \varphi^{-1}(P_i)$ for each i, and order φ^*D as in the proof of Theorem 16. The twisted divisors are $G + jA = (k - j)(\infty) + j(R)$. Then, φ^*C is the matrix-product code $(C_0, C_1, C_2) \cdot \mathbf{V}$ with

$$\mathbf{V} = \left[\zeta^{jm}\right]_{j,m=0}^{2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \alpha^{85} & \alpha^{170} \\ 1 & \alpha^{170} & \alpha^{85} \end{pmatrix} \quad \text{and}$$
$$C_j = \mathbf{v}_j \star C_{\mathcal{L}}(D, G + jA), \quad \mathbf{v}_j := \left(u(\tilde{P}_i)^j\right)_{i=1}^{48}.$$

By Corollary 6, each twist is a scaling of $C_{\mathcal{L}}(D, G_j)$ with $G_j = (k - 1)(\infty) + ([j]R)$. Here,

$[2]R = (\alpha^{51}, \alpha^{65})$, and neither R nor $[2]R$ lies in $\operatorname{supp} D$. By Corollary 17, every twist is a $[48, k, 49-k]_{256}$ MDS code. For instance, for $k = 23$, φ^*C is a $[144, 69, 76]_{256}$ code that glues together three $[48, 23, 26]_{256}$ MDS codes.

Taking instead D to be the sum of the 95 affine points of $\varphi(\mathcal{E}_1(\mathbb{F}_{256}))$, Theorem 11 gives quasi-cyclic $[285, 3k]_{256}$ codes for $1 \le k \le 94$. As $3k \notin \{1, 284\}$, these codes are non-trivial, so they are near-MDS, with minimum distance $285 - 3k$, by the bound $285 > \frac{N+6}{2}$ of [25, Thm. 1.2].

5 Hulls of pullback codes

By Lemma 2, the entanglement needed by an EAQECC built from a classical code is set by the dimension of the code's hull. In this section, we compute the hulls of pullback codes and use them to build quasi-cyclic EAQECCs whose entanglement we choose.

For the rest of this paper, we work in the setting of Theorem 16. For convenience and clarity, we collate our notation.

- $\operatorname{Hull}(C) = C \cap C^\perp$ is the Euclidean hull of a code C.
- $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ is a separable isogeny with $\ker \varphi =: K = \langle T \rangle \subseteq \mathcal{E}_1(\mathbb{F}_q)$.
- $\#K = \ell$ is odd and $\ell \mid q - 1$.
- $\zeta \in \mathbb{F}_q$ is a primitive ℓ^{th} root of unity.
- $u \in \mathbb{F}_q(\mathcal{E}_1)^*$ is a ζ-eigenfunction of $\tau := \tau_T^*$.
- $A \in \operatorname{Div}^0 \mathcal{E}_2$ satisfies $\operatorname{div} u = \varphi^* A$.
- $R = \Sigma(A)$ generates $\ker \hat{\varphi}$.
- $C = C_{\mathcal{L}}(D, G)$ is an $[n, k]_q$ elliptic code on $\mathcal{E}_2$.
- φ^*C is its $[\ell n, \ell k]_q$ pullback code on $\mathcal{E}_1$.
- $G_j = G + jA$ is the twisted divisor. Note that this is now exact (cf. Theorem 16).
- $B_j = C_{\mathcal{L}}(D, G_j)$ is the twisted code (an unscaled version of C_j from earlier).
- Fix $\tilde{P}_i \in \varphi^{-1}(P_i)$ as in the proof of Theorem 16.
- $L_j = u^j \varphi^* \mathcal{L}(G_j)$ is the ζ^j-eigenspace of τ on $\mathcal{L}(\varphi^* G)$.

The standard tool for building EAQECCs is **hull variation** [30, 31]. If C is a self-dual $[n, k, d]_q$ code, scaling $s \le k$ coordinates of an information set by elements λ with $\lambda^2 \ne 1$ gives an equivalent code with hull of dimension $k - s$, so every entanglement between 0 and $n - k$ can be reached. However, applying this procedure directly to a pullback code will generally destroy quasi-cyclicity, as translation by an element of the kernel may send a scaled coordinate to one that has not been scaled. To preserve the quasi-cyclic structure, we restrict our attention to scalings that act uniformly on each fibre.[2]

For $\mathbf{w} = (w_1, \ldots, w_n) \in (\mathbb{F}_q^*)^n$, let $\mathbf{w}^\varphi \in (\mathbb{F}_q^*)^{\ell n}$ take the same value w_i on every coordinate of the fibre $\varphi^{-1}(P_i)$. Translation by T permutes the coordinates inside each fibre, so $\mathbf{w}^\varphi \star \varphi^*C$ is again quasi-cyclic of co-index ℓ, with the same parameters as φ^*C. We call such scalings ***K*-compatible**.

The results are as follows.

- The hull of φ^*C splits along the twists of Theorem 16, pairing twist j with twist $-j$ (Theorem 22).
- Self-orthogonality passes between C and φ^*C in both directions (Theorem 24).
- The residues of one differential produce K-compatible self-dual pullback codes (Proposition 25).
- Starting from these, K-compatible hull variation reaches every multiple of ℓ, together with 0, 1, and 2 when suitable sets of fibres exist (Theorem 26).
- This yields quasi-cyclic EAQECCs, many of them MDS (Corollary 27).

We continue introducing notation that will be used henceforth.

- $\mathbf{v}, \mathbf{w} \in (\mathbb{F}_q^*)^n$ and $\mathbf{v}^\varphi, \mathbf{w}^\varphi \in (\mathbb{F}_q^*)^{\ell n}$ are scaling vectors.
- $C^\natural = \mathbf{w}^\varphi \star \varphi^*C$ is a scaling of φ^*C that is self-orthogonal.
- $C^\sharp = \mathbf{v}^\varphi \star C^\natural$ is a scaling of $C^\natural$ to control its hull dimension.
- $W_j := \{\mathbf{w}^\varphi \star f(\varphi^* D) : f \in L_j\}$ is the image of L_j in $\mathbf{w}^\varphi \star \varphi^*C$.
- Put $m := (\ell - 1)/2$, so that $J := \{-m, \ldots, m\}$ is a set

[2] Note that this is not a necessary condition for preserving quasi-cyclicity.

of representatives of $\mathbb{Z}/\ell\mathbb{Z}$ that is closed under negation.

- For $j \in J$, define the map

$$\begin{aligned} \varepsilon_j : \quad \mathbf{w} \star B_j &\longrightarrow W_j \subseteq \mathbf{w}^\varphi \star \varphi^* C, \\ \mathbf{w} \star g(D) &\longmapsto \mathbf{w}^\varphi \star (u^j \varphi^* g)(\varphi^* D) \end{aligned}$$

which embeds the scaled twisted code into the scaled pullback code.

Since $u^j \varphi^* g \in L_j$ for $g \in \mathcal{L}(G_j)$, the map ε_j is well defined. It is injective because evaluation at D is injective on $\mathcal{L}(G_j)$ when $k < n$. Its image is all of W_j, so ε_j is bijective and $\mathbf{w}^\varphi \star \varphi^* C = \bigoplus_{j \in J} W_j$. As $\mathbf{w}^\varphi$ is constant on fibres, W_j is the ζ^j-eigenspace of translation by T on $\mathbf{w}^\varphi \star \varphi^* C$.

5.1 Hull decomposition

The goal of this subsection is to analyse how the hull of $\varphi^* C$ behaves under K-compatible scalings $\mathbf{w}$. Ling and Sol'e [11, Prop. 4.1] analyse the decomposition of quasi-cyclic codes into constituent codes and the interaction of these constituents with the Euclidean inner product. Lemma 20 and Theorem 22 adapt these ideas to the setting of pullback codes. The resulting decomposition has a particularly simple orthogonality structure, where two eigenspaces are orthogonal unless their indices are negatives of one another.

Lemma 20 (Twisted inner products)

Let $j, j' \in J$.

(a) Let $f = u^j \varphi^* g \in L_j$ and $f' = u^{j'} \varphi^* g' \in L_{j'}$, with $g \in \mathcal{L}(G_j)$ and $g' \in \mathcal{L}(G_{j'})$. Then,

$$\Big(\mathbf{w}^\varphi \star f(\varphi^* D)\Big) \cdot \Big(\mathbf{w}^\varphi \star f'(\varphi^* D)\Big) = \begin{cases} \ell \sum_{i=1}^n w_i^2 \, g(P_i) g'(P_i) & \text{if } j' = -j \\ 0 & \text{otherwise.} \end{cases}$$

(b) $W_j \perp W_{j'}$ unless $j' = -j$.

(c) For all $\mathbf{c} \in \mathbf{w} \star B_j$ and $\mathbf{c}' \in \mathbf{w} \star B_{-j}$, we have $\varepsilon_j(\mathbf{c}) \cdot \varepsilon_{-j}(\mathbf{c}') = \ell \, \mathbf{c} \cdot \mathbf{c}'$.

Proof. For (a), since $\varphi^{-1}(P_i) = \{\tilde{P}_i \oplus [t]T : 0 \le t < \ell\}$ and f and f' are eigenfunctions of τ with eigenvalues ζ^j and $\zeta^{j'}$, we have

$$\begin{aligned} &(\mathbf{w}^\varphi \star f(\varphi^* D)) \cdot (\mathbf{w}^\varphi \star f'(\varphi^* D)) \\ &= \sum_{i=1}^n \sum_{t=0}^{\ell-1} \Big(w_i f(\tilde{P}_i \oplus [t]T)\Big)\Big(w_i f'(\tilde{P}_i \oplus [t]T)\Big) \\ &= \sum_{i=1}^n \sum_{t=0}^{\ell-1} w_i^2 (\tau^t f)(\tilde{P}_i) \cdot (\tau^t f')(\tilde{P}_i) \\ &= \sum_{i=1}^n w_i^2 \sum_{t=0}^{\ell-1} \zeta^{jt} u(\tilde{P}_i)^j g(P_i) \cdot \zeta^{j't} u(\tilde{P}_i)^{j'} g'(P_i) \\ &= \sum_{i=1}^n w_i^2 u(\tilde{P}_i)^{j+j'} g(P_i) g'(P_i) \sum_{t=0}^{\ell-1} \zeta^{(j+j')t}. \end{aligned}$$

The inner sum is ℓ if $\ell \mid j + j'$ and 0 otherwise. As $j, j' \in J$, we have $|j + j'| \le \ell - 1$, so $\ell \mid j + j'$ holds exactly when $j' = -j$, in which case $u(\tilde{P}_i)^{j+j'} = 1$.

(b) follows from (a), as W_j and $W_{j'}$ consist of the vectors $\mathbf{w}^\varphi \star f(\varphi^* D)$ with $f \in L_j$ and $f \in L_{j'}$ respectively.

For (c), write $\mathbf{c} = \mathbf{w} \star g(D)$ and $\mathbf{c}' = \mathbf{w} \star g'(D)$ with $g \in \mathcal{L}(G_j)$ and $g' \in \mathcal{L}(G_{-j})$. Then $\varepsilon_j(\mathbf{c}) = \mathbf{w}^\varphi \star (u^j \varphi^* g)(\varphi^* D)$ and $\varepsilon_{-j}(\mathbf{c}') = \mathbf{w}^\varphi \star (u^{-j} \varphi^* g')(\varphi^* D)$, so (a) gives

$$\varepsilon_j(\mathbf{c}) \cdot \varepsilon_{-j}(\mathbf{c}') = \ell \sum_{i=1}^n w_i g(P_i) \cdot w_i g'(P_i) = \ell \, \mathbf{c} \cdot \mathbf{c}'. \quad \square$$

Theorem 22(b) needs another lemma from linear algebra.

Lemma 21 (Cross-orthogonal dimension)

Let $U, V \subseteq \mathbb{F}_q^n$ be subspaces with $\dim U = \dim V = k$. Then

$$\dim(U \cap V^\perp) = \dim(V \cap U^\perp).$$

Proof. Let $\mathbf{G}_U, \mathbf{G}_V \in \mathbb{F}_q^{k \times n}$ be (row-wise) generator matrices for U and V respectively. The map $\mathbb{F}_q^k \to U$, $\mathbf{m} \mapsto \mathbf{m}\mathbf{G}_U$, is an isomorphism. A vector $\mathbf{m}\mathbf{G}_U \in U$ lies in $V^\perp$ if and only if it is orthogonal to every row of $\mathbf{G}_V$, that is, $\mathbf{m}\mathbf{G}_U \mathbf{G}_V^\top = \mathbf{0}$. Hence, $U \cap V^\perp \cong \ker \mathbf{G}_U \mathbf{G}_V^\top$ has dimension $k - \operatorname{rank}(\mathbf{G}_U \mathbf{G}_V^\top)$. The claim follows from

$$\operatorname{rank}(\mathbf{G}_U \mathbf{G}_V^\top) = \operatorname{rank}\Big((\mathbf{G}_U \mathbf{G}_V^\top)^\top\Big) = \operatorname{rank}(\mathbf{G}_V \mathbf{G}_U^\top). \quad \square$$

Theorem 22 specialises to pullback codes the description of duals of quasi-cyclic codes through their constituents [11, Prop. 4.1, Thm. 4.2]. The genus-1 contributions include a geometric meaning for the $j \leftrightarrow -j$ pairing, and that each constituent is an explicit elliptic code B_j, so the **cross-hulls** H_j can be

computed with the group law.

Theorem 22 (Hull decomposition)

For $j \in J$, let $H_j := (\mathbf{w} \star B_j) \cap (\mathbf{w} \star B_{-j})^{\perp}$ and $h_j := \dim H_j$. Then,

(a) $\mathrm{Hull}(\mathbf{w}^{\varphi} \star \varphi^* C) = \bigoplus_{j \in J} \varepsilon_j(H_j)$;

(b) $h_{-j} = h_j$ and $h_0 = \dim \mathrm{Hull}(\mathbf{w} \star C)$, so

$$\dim \mathrm{Hull}(\mathbf{w}^{\varphi} \star \varphi^* C) = h_0 + 2\sum_{j=1}^{m} h_j;$$

(c) $\mathrm{Hull}(\mathbf{w}^{\varphi} \star \varphi^* C)$ is K-stable, and its K-fixed part is the ℓ-fold repetition of $\mathrm{Hull}(\mathbf{w} \star C)$;

(d) if $\mathbf{w}^{\varphi} \star \varphi^* C$ is LCD, then so is $\mathbf{w} \star C$. In general, $\dim \mathrm{Hull}(\mathbf{w}^{\varphi} \star \varphi^* C) \geq \dim \mathrm{Hull}(\mathbf{w} \star C)$, and the two dimensions have the same parity.

Proof. For (a), let $\mathbf{x} \in \mathbf{w}^{\varphi} \star \varphi^* C$. Since $\mathbf{w}^{\varphi} \star \varphi^* C = \bigoplus_{j \in J} W_j$, we may write $\mathbf{x} = \sum_{j \in J} \mathbf{x}_j$ with $\mathbf{x}_j \in W_j$. Since ε_j is surjective, we may write $\mathbf{x}_j = \varepsilon_j(\mathbf{c}_j)$ with $\mathbf{c}_j \in \mathbf{w} \star B_j$. By Lemma 20(b), $W_j \perp W_{j'}$ unless $j' = -j$. Using Lemma 20(c) for the fifth equivalence, we obtain

$$\begin{aligned}
& \mathbf{x} \in \mathrm{Hull}(\mathbf{w}^{\varphi} \star \varphi^* C) \\
\iff\ & \mathbf{x} \perp \mathbf{w}^{\varphi} \star \varphi^* C \\
\iff\ & \mathbf{x} \perp W_j \quad \forall\, j \in J \\
\iff\ & \mathbf{x}_{-j} \perp W_j \quad \forall\, j \in J \\
\iff\ & \varepsilon_{-j}(\mathbf{c}_{-j}) \cdot \varepsilon_j(\mathbf{c}') = 0 \quad \forall\, j \in J \quad \forall\, \mathbf{c}' \in \mathbf{w} \star B_j \\
\iff\ & \ell\, \mathbf{c}_{-j} \cdot \mathbf{c}' = 0 \quad \forall\, j \in J \quad \forall\, \mathbf{c}' \in \mathbf{w} \star B_j \\
\iff\ & \mathbf{c}_{-j} \perp \mathbf{w} \star B_j \quad \forall\, j \in J \\
\iff\ & \mathbf{c}_{-j} \in (\mathbf{w} \star B_{-j}) \cap (\mathbf{w} \star B_j)^{\perp} = H_{-j} \quad \forall\, j \in J \\
\iff\ & \mathbf{x}_{-j} \in \varepsilon_{-j}(H_{-j}) \quad \forall\, j \in J \\
\iff\ & \mathbf{x} \in \bigoplus_{j \in J} \varepsilon_j(H_j).
\end{aligned}$$

This proves (a). For (b), $h_j = h_{-j}$ follows from Lemma 21 with $U = \mathbf{w} \star B_j$ and $V = \mathbf{w} \star B_{-j}$. Also, $H_0 = \mathrm{Hull}(\mathbf{w} \star C)$ because $G_0 = G$.

For (c), translation by T is a permutation automorphism of $\mathbf{w}^{\varphi} \star \varphi^* C$, hence of its dual and hull. As $\mathbf{w}^{\varphi}$ is constant on fibres, Proposition 13 shows that the K-fixed part of $\mathbf{w}^{\varphi} \star \varphi^* C$ is W_0. So the K-fixed part of the hull is $W_0 \cap (\mathbf{w}^{\varphi} \star \varphi^* C)^{\perp} = \varepsilon_0(H_0)$ by (a), and ε_0 is the ℓ-fold repetition map because $u^0 = 1$. Since $H_0 = \mathrm{Hull}(\mathbf{w} \star C)$, the claim follows.

(d) follows from the dimension formula in (b). □

5.2 Self-orthogonality and self-duality

We begin with a sufficient condition for $\mathbf{w} \star C$ to be Euclidean self-orthogonal, extending [32, Lem. 3–4].

Lemma 23 (Self-orthogonal scalings [32])

Let $\eta \in \Omega_{\mathcal{E}}$ with $\operatorname{div} \eta \geq 2G - D$, and put $\mathbf{r} := (\mathrm{Res}_{P_i}(\eta))_{i=1}^{n}$.

(a) $\mathbf{r} \star C \subseteq C^{\perp}$, with equality if $\operatorname{div} \eta = 2G - D$.

(b) If $\mathbf{r} = \lambda\, \mathbf{w} \star \mathbf{w}$ for some $\lambda \in \mathbb{F}_q^*$, then $\mathbf{w} \star C$ is self-orthogonal (in particular, self-dual if $\operatorname{div} \eta = 2G - D$).

Proof. (a) This is [32, Lem. 3–4].

(b) For all $\mathbf{c}, \mathbf{c}' \in C$, (a) gives

$$(\mathbf{w} \star \mathbf{c}) \cdot (\mathbf{w} \star \mathbf{c}') = \sum_{i=1}^{n} w_i^2 c_i c_i' = \lambda^{-1} (\mathbf{r} \star \mathbf{c}) \cdot \mathbf{c}' = 0,$$

so $\mathbf{w} \star C$ is self-orthogonal. If $\operatorname{div} \eta = 2G - D$, then by (a), $(\mathbf{w} \star C)^{\perp} = \mathbf{w}^{-1} \star C^{\perp} = \mathbf{w}^{-1} \star \mathbf{r} \star C = \lambda\, \mathbf{w} \star C = \mathbf{w} \star C$. □

Next, we show that self-orthogonality passes between $\mathbf{w} \star C$ and its pullback $\mathbf{w}^{\varphi} \star \varphi^* C$. Recall that $\mathbf{w}^{\varphi} \star \varphi^* C$ is self-orthogonal if and only if $\mathrm{Hull}(\mathbf{w}^{\varphi} \star \varphi^* C) = \mathbf{w}^{\varphi} \star \varphi^* C$.

Theorem 24 (Self-orthogonality transfer)

Let $k \geq 3$ and $\mathbf{w} \in (\mathbb{F}_q^*)^n$. Then, $\mathbf{w}^{\varphi} \star \varphi^* C$ is self-orthogonal if and only if $\mathbf{w} \star C$ is self-orthogonal.

Proof. Put $\lambda_{\mathbf{w}}(f) := \sum_{i=1}^{n} w_i^2 f(P_i)$ for $f \in \mathcal{L}(2G)$. By Lemma 20(a), the inner products in $\mathbf{w}^{\varphi} \star \varphi^* C$ are 0 or $\ell \lambda_{\mathbf{w}}(gg')$, where $u^j \varphi^* g \in L_j$, $u^{-j} \varphi^* g' \in L_{-j}$, and

$$gg' \in \mathcal{L}(G_j) \cdot \mathcal{L}(G_{-j}) \subseteq \mathcal{L}(G_j + G_{-j}) = \mathcal{L}(2G).$$

If $\mathbf{w}^{\varphi} \star \varphi^* C$ is self-orthogonal, then so is the subcode $W_0 = \varepsilon_0(\mathbf{w} \star C)$. By Lemma 20(c) with $j = 0$, and as $\ell \neq 0$ in $\mathbb{F}_q$, $\mathbf{w} \star C$ is self-orthogonal too. Conversely, suppose $\mathbf{w} \star C$ is self-orthogonal. Then

$$0 = (\mathbf{w} \star h(D)) \cdot (\mathbf{w} \star h'(D)) = \lambda_{\mathbf{w}}(hh') \quad \forall\, h, h' \in \mathcal{L}(G),$$

so $\lambda_{\mathbf{w}}$ vanishes on $\mathcal{L}(G) \cdot \mathcal{L}(G)$, which is $\mathcal{L}(2G)$ by

[33, Thm. 6]. In particular, $\lambda_{\mathbf{w}}(gg') = 0$ for all $j \in J$, $g \in \mathcal{L}(G_j)$, and $g' \in \mathcal{L}(G_{-j})$, since $gg' \in \mathcal{L}(2G)$. By Lemma 20(a), every inner product in $\mathbf{w}^{\varphi} \star \varphi^* C$ therefore vanishes, proving self-orthogonality of $\mathbf{w}^{\varphi} \star \varphi^* C$. □

Suppose $k \geq 3$ and $\mathbf{w} \star C$ is self-orthogonal (for instance, when Lemma 23(b) applies). By Theorem 24, $\mathbf{w}^{\varphi} \star \varphi^* C$ is self-orthogonal too. The CSS construction [34] (the case $c = 0$ of Lemma 2) then gives a quasi-cyclic $[\![\ell n, \ell(n-2k), \geq \ell k; 0]\!]_q$ code.

The next result makes Corollary 12 explicit by identifying the scaling vectors $\mathbf{r}$ and $\mathbf{r}^{\varphi}$ of the iso-duality. If, moreover, the entries of $\mathbf{r}$ are all quadratic residues or all nonresidues, we obtain a self-dual pullback code.

Proposition 25 (Self-dual pullback codes)

Suppose that $D - 2G$ is principal. Then, there exists $\omega \in \Omega_{\mathcal{E}_2}$ with $\operatorname{div}\omega = 2G - D$. Put $\mathbf{r} := (\operatorname{Res}_{P_i}\omega)_{i=1}^n \in (\mathbb{F}_q^*)^n$. Then

$$C^{\perp} = \mathbf{r} \star C \quad \text{and} \quad (\varphi^* C)^{\perp} = \mathbf{r}^{\varphi} \star \varphi^* C.$$

Moreover, if $\mathbf{r} = \lambda\, \mathbf{w} \star \mathbf{w}$ for some $\lambda \in \mathbb{F}_q^*$ and $\mathbf{w} \in (\mathbb{F}_q^*)^n$, then $\mathbf{w} \star C$ and $\mathbf{w}^{\varphi} \star \varphi^* C$ are self-dual. This holds automatically if q is even.

Proof. Write $D - 2G = \operatorname{div} h$. As canonical divisors on $\mathcal{E}_2$ are principal (Section 2.2), $\omega := h^{-1}\omega_{\mathcal{E}_2}$ has $\operatorname{div}\omega = 2G - D$, where $\omega_{\mathcal{E}_2}$ is the invariant differential. The claims for C then follow from Lemma 23(a) and (b).

As φ is separable, it is unramified, so the different of $\mathbb{F}_q(\mathcal{E}_1)/\varphi^*\mathbb{F}_q(\mathcal{E}_2)$ is zero [3, §3.4]. Hence, $\operatorname{div}\varphi^*\omega = \varphi^* \operatorname{div}\omega = 2\varphi^* G - \varphi^* D$. If t is a uniformiser at P_i, then $\varphi^* t$ is a uniformiser at each $P \in \varphi^{-1}(P_i)$, giving $\operatorname{Res}_P(\varphi^*\omega) = \operatorname{Res}_{P_i}(\omega) = r_i$. Applying Lemma 23 to $\varphi^* C$ with $\eta = \varphi^*\omega$, whose residue vector is $\mathbf{r}^{\varphi}$, gives the claims for $\varphi^* C$. □

5.3 Quasi-cyclic hull variation

Starting from a self-orthogonal pullback code $C^{\natural} := \mathbf{w}^{\varphi} \star \varphi^* C$ (such as from Theorem 24 or Proposition 25), we now show how K-compatible scalings lower its hull. First, we choose the fibres to be scaled. Number the fibres from 1 to n, choose $1 \leq s \leq k$ of them to be scaled, and represent them by an indexing set $I \subseteq \{1, \ldots, n\}$. Write $D_I := \sum_{i \in I}(P_i)$ for the subdivisor of D indexed by I, and I^c for the complement of I in $\{1, \ldots, n\}$.

Next, we choose the scaling $\mathbf{v} \in (\mathbb{F}_q^*)^n$. For each $i \in I$, choose $\lambda_i \in \mathbb{F}_q^*$ with $\lambda_i^2 \neq 1$. Put $v_i := \lambda_i$ for $i \in I$ and $v_i := 1$ otherwise, and write $C^{\sharp} := \mathbf{v}^{\varphi} \star C^{\natural}$. The code $C^{\sharp}$ depends on I and the λ_i, but we suppress this in the notation. Theorem 26 states that the hull dimension drops by ℓ for each fibre scaled, up to $k-1$ fibres. The case $s < k$ follows from our generalisation of [35, Thm. 1(i)] from genus 0 to 1, but the case $s = k$ exhibits intricate behaviour due to the elliptic group law. We study this special case using the hull decomposition in Theorem 22.

Theorem 26 (Quasi-cyclic hull variation)

Let $C^{\natural}$, I, λ_i, and $C^{\sharp}$ be as above. Then

(a) $C^{\sharp}$ is quasi-cyclic of co-index ℓ;

(b) if $s < k$, then $\dim \operatorname{Hull}(C^{\sharp}) = \ell(k - s)$;

(c) suppose $s = k = \frac{n}{2}$ and $C^{\natural}$ is self-dual. Then, $\dim \operatorname{Hull}(C^{\sharp})$ is

$$\begin{cases} 0 & \text{if } \Sigma(D_I) \ominus \Sigma(G) \notin \ker \hat{\varphi}, \\ 2 & \text{if } \Sigma(D_I) \ominus \Sigma(G) \in \ker \hat{\varphi} \setminus \{\infty\}, \\ 1 \text{ or } 2 & \text{if } \Sigma(D_I) = \Sigma(G). \end{cases}$$

In the last case, the dimension is 2 if and only if

$$\sum_{i \in I} \frac{w_i^2 g(P_i)^2}{\lambda_i^2 - 1} = 0, \tag{3}$$

where g spans $\mathcal{L}(G - D_{I^c})$. In particular, it is 2 when all λ_i are equal.

Proof. (a) The scaling $\mathbf{v}^{\varphi} \star \mathbf{w}^{\varphi}$ is constant on fibres (i.e. K-compatible), so $C^{\sharp}$ is quasi-cyclic.

For the remaining parts, put $\mathbf{a} := \mathbf{v} \star \mathbf{w}$, so that $C^{\sharp} = \mathbf{a}^{\varphi} \star \varphi^* C$. For $j \in J$, consider the evaluation map

$$\operatorname{ev}_j : \mathcal{L}(G_j) \to \mathbb{F}_q^I, \qquad g \mapsto (g(P_i))_{i \in I}.$$

Its kernel is $\mathcal{L}(G_j - D_I)$. Define $\boldsymbol{\mu} \in (\mathbb{F}_q^*)^I$ by $\mu_i := (\lambda_i^2 - 1) w_i^2 \neq 0$ for $i \in I$, and define the nondegenerate bilinear form $\beta(\mathbf{x}, \mathbf{y}) := \mathbf{x} \cdot (\boldsymbol{\mu} \star \mathbf{y}) = \sum_{i \in I} \mu_i x_i y_i$ on $\mathbb{F}_q^I$. Write $\mathbf{x} \perp_{\beta} \mathbf{y}$ to denote $\beta(\mathbf{x}, \mathbf{y}) = 0$.

We apply Theorem 22 with $\mathbf{a}$ in place of $\mathbf{w}$, so that

hull membership reduces to membership of the cross-hulls $H_j = (\mathbf{a}\star B_j)\cap(\mathbf{a}\star B_{-j})^{\perp}$. Let $\mathbf{a}\star g(D)\in \mathbf{a}\star B_j$ be arbitrary, where $g\in\mathcal{L}(G_j)$. Then, $\mathbf{a}\star g(D)\in H_j$ if and only if $(\mathbf{a}\star g(D))\perp(\mathbf{a}\star g'(D))$ for all $g'\in\mathcal{L}(G_{-j})$. Given such a g', write $c_i := w_i^2 g(P_i)g'(P_i)$ for $1\le i\le n$. As $C^{\natural}$ is self-orthogonal, Lemma 20(a) gives $\sum_{i=1}^n c_i = 0$. As $v_i = 1$ for $i\in I^{\mathrm{c}}$, we have

$$\begin{aligned}0 &= \big(\mathbf{a}\star g(D)\big)\cdot\big(\mathbf{a}\star g'(D)\big) = \sum_{i=1}^n v_i^2 c_i - 0\\ &= \sum_{i=1}^n v_i^2 c_i - \sum_{i=1}^n c_i = \sum_{i\in I\sqcup I^{\mathrm{c}}}(v_i^2-1)c_i\\ &= \sum_{i\in I}(\lambda_i^2-1)c_i = \beta\big(\mathrm{ev}_j(g),\mathrm{ev}_{-j}(g')\big).\end{aligned}$$

Hence, $\mathbf{a}\star g(D)\in H_j$ if and only if $\mathrm{ev}_j(g)\perp_\beta \mathrm{im}(\mathrm{ev}_{-j})$.

(b) When $s<k$, the Riemann–Roch theorem gives

$$\begin{aligned}\dim\ker(\mathrm{ev}_{-j}) &= \deg(G_{-j}-D_I) = k-s>0, \quad \text{so}\\ \mathrm{rank}(\mathrm{ev}_{-j}) &= k-(k-s) = s = \dim\mathbb{F}_q^I\end{aligned}$$

and ev_{-j} is surjective. Hence,

$$\begin{aligned}\mathbf{a}\star g(D)\in H_j \quad&\Longleftrightarrow\quad \mathrm{ev}_j(g)\perp_\beta \mathrm{im}(\mathrm{ev}_{-j})\\ \Longleftrightarrow\quad \mathrm{ev}_j(g)\perp_\beta \mathbb{F}_q^I \quad&\Longleftrightarrow\quad \mathrm{ev}_j(g)=0\end{aligned}$$

since β is nondegenerate. Thus,

$$H_j = \{\mathbf{a}\star g(D) : g\in\ker(\mathrm{ev}_j)\} \quad\text{and}\quad h_j = k-s.$$

By Theorem 22(b), $\dim\mathrm{Hull}(C^{\sharp}) = \sum_{j\in J} h_j = \ell(k-s)$.

(c) Now $s=k$, so each ev_j is a map between spaces of dimension k. Since $G_j - D_I$ has degree 0, $\ker(\mathrm{ev}_j) = \mathcal{L}(G_j-D_I)$ has dimension 1 if and only if G_j-D_I is principal. By Lemma 3 and Theorem 16, this holds if and only if $\Sigma(D_I)=\Sigma(G_j)=\Sigma(G)\oplus[j]R$, or equivalently, $[j]R = \Sigma(D_I)\ominus\Sigma(G)$. As R has order ℓ, at most one $j\in J$ satisfies this.

Suppose first that $\Sigma(D_I)\ominus\Sigma(G)\notin\ker\hat{\varphi} = \langle R\rangle$. Then, every ev_j is bijective, so every $h_j = 0$ as in (b). Thus, $C^{\sharp}$ is LCD by Theorem 22(a).

Suppose next that $\Sigma(D_I)\ominus\Sigma(G) = [j_0]R$ with $j_0\neq 0$. Then, ev_j is bijective for $j\neq j_0$, so $h_j = 0$ for $j\notin\{\pm j_0\}$. As ev_{-j_0} is surjective and β is nondegenerate, the argument in (b) gives $\mathbf{a}\star g(D)\in H_{j_0}$ if and only if $\mathrm{ev}_{j_0}(g)=0$. This gives $h_{j_0} = \dim\ker(\mathrm{ev}_{j_0}) = 1$. By Theorem 22(b), $h_{-j_0}=1$ and $\dim\mathrm{Hull}(C^{\sharp})=2$.

Suppose finally that $\Sigma(D_I)=\Sigma(G)$. Then, ev_j is bijective for $j\neq 0$, so $h_j=0$ for $j\neq 0$ and $\dim\mathrm{Hull}(C^{\sharp}) = h_0$. Let $V := \mathrm{im}(\mathrm{ev}_0)$, which is a hyperplane in $\mathbb{F}_q^I$ since $\dim\ker(\mathrm{ev}_0)=1$, and let $\mathbf{b}$ span $V^{\perp}\subseteq\mathbb{F}_q^I$.

We first describe $\mathbf{b}$. As $C^{\natural}$ is self-orthogonal, so is $\mathbf{w}\star C$, by the forward direction of Theorem 24 (which does not need $k\ge 3$). As $\dim(\mathbf{w}\star C) = k = \frac{n}{2}$, $\mathbf{w}\star C$ is self-dual. Define $\mathbf{y}\in\mathbb{F}_q^n$ by $y_i := b_i/w_i$ for $i\in I$ and $y_i := 0$ for $i\in I^{\mathrm{c}}$. Since

$$\mathbf{y}\cdot(\mathbf{w}\star g'(D)) = \mathbf{b}\cdot\mathrm{ev}_0(g') = 0 \quad \forall\, g'\in\mathcal{L}(G),$$

it follows that $\mathbf{y}\in(\mathbf{w}\star C)^{\perp} = \mathbf{w}\star C$, i.e. $\mathbf{y} = \mathbf{w}\star g(D)$ for some $g\in\mathcal{L}(G)^*$. As $\mathbf{y}$ vanishes on I^{c}, we have $\mathrm{div}\,g\ge D_{I^{\mathrm{c}}} - G$. Both divisors have degree $0 = (n-s)-k$, so $\mathrm{div}\,g = D_{I^{\mathrm{c}}}-G$, i.e. g is nonzero at each P_i, $i\in I$. Thus, $\mathbf{b} = (w_i y_i)_{i\in I} = \big(w_i^2 g(P_i)\big)_{i\in I}\in(\mathbb{F}_q^*)^I$.

By the criterion before the proof of (b) with $j=0$,

$$\begin{aligned}H_0 &= \{\mathbf{a}\star g(D) : \mathrm{ev}_0(g)\in V\cap V^{\perp_\beta}\}, \quad \text{so}\\ \dim H_0 &= \dim\ker(\mathrm{ev}_0) + \dim(V\cap V^{\perp_\beta})\\ h_0 &= 1+\dim(V\cap V^{\perp_\beta}).\end{aligned}$$

But

$$\begin{aligned}V^{\perp_\beta} &= \{\mathbf{y} : V\cdot(\boldsymbol{\mu}\star\mathbf{y})=0\} = \{\mathbf{y} : \boldsymbol{\mu}\star\mathbf{y}\in V^{\perp}\}\\ &= \boldsymbol{\mu}^{-1}\star V = \langle\boldsymbol{\mu}^{-1}\star\mathbf{b}\rangle, \quad \text{so}\end{aligned}$$

$$\begin{aligned}h_0 = 2 \quad&\Longleftrightarrow\quad \dim(V\cap V^{\perp_\beta}) = 1\\ &\Longleftrightarrow\quad \boldsymbol{\mu}^{-1}\star\mathbf{b}\in V = \langle\mathbf{b}\rangle^{\perp}\\ &\Longleftrightarrow\quad (\boldsymbol{\mu}^{-1}\star\mathbf{b})\cdot\mathbf{b} = \sum_{i\in I}\frac{w_i^2 g(P_i)^2}{\lambda_i^2-1} = 0,\end{aligned}$$

which is Equation (3). Otherwise, $h_0 = 1$.

If all λ_i equal λ, the sum is $(\lambda^2-1)^{-1}(\mathbf{w}\star g(D))\cdot(\mathbf{w}\star g(D)) = 0$, so $h_0 = 2$. □

With the aid of Theorem 22(a), the proof of Theorem 26 also locates the hull explicitly. Let W_j and ε_j be defined with $\mathbf{v}\star\mathbf{w}$ in place of $\mathbf{w}$. In the second case of (c), the hull is $\varepsilon_{j_0}(H_{j_0})\oplus\varepsilon_{-j_0}(H_{-j_0})\subseteq W_{j_0}\oplus W_{-j_0}$. In the third case, it is $\varepsilon_0(H_0)\subseteq W_0$, so it is a repetition code. In the first case, every ev_j is bijective, so φ^*D_I is an information set of φ^*C.

6 Quasi-cyclic EAQECCs

Combining Theorem 26 with Lemma 2, we obtain EAQECCs whose classical codes are quasi-cyclic, with a prescribed amount of entanglement. We call such EAQECCs **quasi-cyclic** as the check matrices inherit the block-circulant structure.

Physically, this structure is a symmetry of the qudits. Group the sender's ℓn qudits into n blocks of ℓ, one for each fibre $\varphi^{-1}(P_i)$. Translation by T induces a permutation π (and hence, a unitary operator U_π) that cycles the qudits inside each block. The stabiliser group on the sender's qudits (resp. its commuting subspace) is generated by $X(\mathbf{c})$ and $Z(\mathbf{c})$, for $\mathbf{c} \in (C^\sharp)^\perp$ (resp. $\mathbf{c} \in \mathrm{Hull}(C^\sharp)$). As $(C^\sharp)^\perp$ (resp. $\mathrm{Hull}(C^\sharp)$) is π-invariant, conjugation by U_π merely permutes these generators. We leave a full description of quasi-cyclic EAQECCs, including the behaviour of the receiver's qudits under this symmetry, to future work.

Corollary 27 (LCD and quasi-cyclic EAQECCs)

Suppose that $C^\natural = \mathbf{w}^\varphi \star \varphi^* C$ is self-dual (for instance, as in Proposition 25), so that $n = 2k$. Assume also that $q \geq 4$. Let $d := d(\varphi^* C) \in \{\ell k, \ell k + 1\}$.

(a) For $\#I = k$, the code $C^\sharp$ is LCD if and only if $\Sigma(D_I) \ominus \Sigma(G) \notin \ker \hat{\varphi}$, which is automatic if $\varphi^* C$ is MDS.

(b) For $1 \leq s \leq k-1$, there exists a quasi-cyclic EAQECC with parameters $[\![2\ell k,\ \ell s,\ d;\ \ell s]\!]_q$. If some k-subset I satisfies the condition in (a), the same holds for $s = k$.

(c) If $\varphi^* C$ is MDS, the codes in (b) are $[\![2\ell k,\ \ell s,\ \ell k + 1;\ \ell s]\!]_q$ MDS EAQECCs for all $1 \leq s \leq k$.

(d) If some k-subset I has $\Sigma(D_I) \ominus \Sigma(G) \in \ker \hat{\varphi}$, there exists a quasi-cyclic EAQECC with parameters $[\![2\ell k,\ \ell k - 2,\ d;\ \ell k - 2]\!]_q$.

(e) If some k-subset I has $\Sigma(D_I) = \Sigma(G)$, and $q \geq 7$ when q is odd, there exists a quasi-cyclic EAQECC with parameters $[\![2\ell k,\ \ell k - 1,\ d;\ \ell k - 1]\!]_q$.

More generally, if $C^\natural$ is self-orthogonal, so that $n \geq 2k$, then Theorem 26(b) gives quasi-cyclic $[\![\ell n,\ \ell s,\ d(\varphi^* C);\ \ell(n - 2k + s)]\!]_q$ codes for $0 \leq s \leq k - 1$.

Proof. The bounds on q give $\lambda \in \mathbb{F}_q^*$ with $\lambda^2 \neq 1$.

(a) Apply Theorem 26(c), together with Corollary 17 in the MDS case.

(b) Applying Lemma 2 to the $[2\ell k, \ell k, d]$ code $C^\sharp$ with hull dimension h gives $[\![2\ell k,\ \ell k - h,\ d;\ \ell k - h]\!]_q$, where $h = \ell(k - s)$ by Theorem 26(b).

(c) Apply the MDS clause of Lemma 2, which applies since $\dim(\varphi^* C) = \ell k = \frac{2\ell k}{2}$.

(d) Apply Theorem 26(c) with $\lambda_i = \lambda$ for every $i \in I$ to obtain hull dimension 2.

(e) Put $\lambda_i = \lambda$ for $i \in I \setminus \{i_0\}$ and $\lambda_{i_0} = \lambda' \in \mathbb{F}_q^*$ with $1, \lambda^2, \lambda'^2$ distinct, which is possible as $\mathbb{F}_q^*$ has at least three squares. Adopting the notation of Theorem 26(c), writing $c_i := w_i^2 g(P_i)^2$, and using $\sum_{i \in I} c_i = 0$ (self-orthogonality of $\mathbf{w} \star C$, as g vanishes at P_i for $i \in I^c$), the left side of Equation (3) then collapses to

$$\sum_{i \in I} \frac{c_i}{\lambda_i^2 - 1} = \frac{1}{\lambda^2 - 1} \sum_{i \neq i_0} c_i + \frac{1}{\lambda'^2 - 1} c_{i_0} = c_{i_0} \left((\lambda'^2 - 1)^{-1} - (\lambda^2 - 1)^{-1} \right).$$

Each factor is nonzero, so Equation (3) fails and the hull dimension is 1. The last claim follows from Lemma 2 in the same way. □

Example 28 (QC MDS EAQECCs over $\mathbb{F}_{2^8}$ [29])

Take $k = 23$ in Example 19. Remove from D the two points with $x = \alpha^5$, and call the result D'. Then, D' has $n = 46 = 2k$ points. As $\operatorname{supp} D'$ is closed under negation $\ominus$ and contains no point of order 2, its points cancel in pairs, so $\Sigma(D') = \infty = \Sigma(2G)$, hence $D' - 2G$ is principal. Applying Construction 18 to D' gives the quasi-cyclic $[138, 69, 70]_{256}$ MDS code $\varphi^* C = C_{\mathcal{L}}(\varphi^* D', \varphi^* G)$ of co-index 3, where $C = C_{\mathcal{L}}(D', 23(\infty))$.

As q is even, Proposition 25 gives $\mathbf{w} \in (\mathbb{F}_{256}^*)^{46}$ such that $\mathbf{w}^\varphi \star \varphi^* C$ is self-dual. By Theorem 26, scaling s fibres by $\lambda = \alpha$ gives hull dimension $3(23 - s)$ for $s = 0, \ldots, 23$. Corollary 27 then gives quasi-cyclic $[\![138,\ 3s,\ 70;\ 3s]\!]_{256}$ MDS EAQECCs for each $1 \leq s \leq 23$.

Characteristic 2 avoids two obstructions over fields of odd characteristic. First, the only self-dual codes over $\mathbb{F}_q$ with $q \equiv 3 \pmod 4$ have

lengths divisible by 4. Second, the residues of Proposition 25 must all lie in one square class, which rarely happens for long D.

Example 29 (QC MDS CSS codes over $\mathbb{F}_{2^8}$)

The pullback codes of Example 19 also give quantum codes without entanglement via the CSS construction following Theorem 24. Let D', C, and $\mathbf{w}$ be as in Example 28, so that $\mathbf{w} \star C$ is self-dual. For odd $3 \le k' \le 21$, put $C' := C_{\mathcal{L}}(D', k'(\infty))$. Then, $\mathbf{w} \star C'$ is a subcode of the self-dual code $\mathbf{w} \star C$, so it is self-orthogonal. By Theorem 24, $\mathbf{w}^{\varphi} \star \varphi^* C'$ is self-orthogonal too. By Construction 18, $\varphi^* C'$ is a $[138,\ 3k',\ 139 - 3k']_{256}$ MDS code, so its dual is a $[138,\ 138 - 3k',\ 3k' + 1]_{256}$ MDS code. The CSS construction then gives quasi-cyclic $[\![138,\ 138 - 6k',\ 3k' + 1]\!]_{256}$ MDS codes.

For instance, $k' = 11$ gives a quasi-cyclic $[\![138, 72, 34]\!]_{256}$ quantum MDS code. Quantum MDS codes with these parameters have already been obtained from Euclidean self-orthogonal GRS codes [36, Thm. 5.3], but they carry no quasi-cyclic structure.

Example 30 (Beyond length $q+1$ over $\mathbb{F}_{64}$)

Let $\mathbb{F}_{64} = \mathbb{F}_2(\alpha)$ with $\alpha^6 = \alpha + 1$, and $\mathcal{E}_1 : y^2 + xy = x^3 + \beta$ with $\beta = \alpha^3 + \alpha^2 + \alpha + 1$. Then, $N = 72$ and $\mathcal{E}_1(\mathbb{F}_{64})$ has rational 3-torsion. Take

- $T = (\alpha^2, \alpha) \in \mathcal{E}_1$ of order $\ell = 3$,
- $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ the 3-isogeny with kernel $\langle T \rangle$,
- $Q = (0, \alpha^9) \in \mathcal{E}_1$ of order 2,
- D the sum of the 22 points of $\varphi(\mathcal{E}_1(\mathbb{F}_{64}))$ excluding ∞ and $\varphi(Q)$, and
- $G = 11(\infty)$.

Then, $\operatorname{supp} D$ is closed under negation $\ominus$ and contains no point of order 2, as $\varphi(Q)$ is the only such point of $\mathcal{E}_2$. Hence, $\Sigma(D) = \infty = \Sigma(2G)$, so $D - 2G$ is principal, and $\varphi^* C$ is a quasi-cyclic $[66, 33, 33]_{64}$ code. Its length exceeds $q + 1 = 65$, the length of the longest GRS codes over $\mathbb{F}_{64}$.

As q is even, $\mathbf{w}^{\varphi} \star \varphi^* C$ is self-dual. For $s < 11$, the hull dimensions are $3(11 - s)$. For $s = 11$, all three cases of Theorem 26(c) occur [37], giving quasi-cyclic EAQECCs

$$[\![66,\ c,\ 33;\ c]\!]_{64}, \quad c \in \{3, 6, \ldots, 30, 31, 32, 33\}.$$

For comparison, the same parameters follow from the hull-variation theorem of [30] applied to the self-dual near-MDS elliptic codes of [9], but without quasi-cyclicity.

There are other scalings $\mathbf{v}$ that preserve quasi-cyclicity but are not constant on fibres. One example is given by $v_{P \oplus T} = \zeta^b v_P$, which changes how the twists are paired. It is also natural to search for a Hermitian analogue of Section 5, as the Hermitian construction of quantum codes yields better parameters. In the Hermitian case, if $\ell \mid q + 1$ then $\zeta^q = \zeta^{-1}$, so each twist is paired with itself rather than with its negative. We leave these directions to future work.

7 Conclusion

We introduced the notion of pullback codes, which are constructed by pulling back an elliptic code C on $\mathcal{E}_2$ to a code $\varphi^* C$ on $\mathcal{E}_1$ via a separable cyclic isogeny $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$. This construction multiplies the code length by a factor of ℓ, while preserving its rate and relative designed distance. Moreover, the kernel translations endow the pullback code with a quasi-cyclic structure of co-index ℓ. When $\ell \mid q - 1$, the code admits a matrix-product decomposition into twists of the original code. This decomposition, together with the index-two construction [7], provides quasi-cyclic MDS elliptic codes. Finally, the hull decomposes according to the same constituent codes. By scaling entire fibres, we obtain quasi-cyclic EAQECCs with prescribed entanglement, subject to a subset-sum condition on $\mathcal{E}_2$.

To our knowledge, this is the first use of isogenies in the construction of codes. Some natural next steps include

- efficient decoding of $\varphi^* C$ through its twists,
- hull variation using scalings that are not constant on fibres,
- hull variation using Hermitian analogues of Sec-

tion 5,

- a comprehensive study of quasi-cyclic EAQECCs, together with physical and mathematical descriptions,
- an extension to non-cyclic isogenies, and
- analogues for isogenies of higher-dimensional abelian varieties, such as Jacobians of hyperelliptic curves.

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AI declaration

Claude Opus 5.5 was consulted to generalise the proofs of [11, Prop. 4.1, Thm. 4.2], [32, Lem. 3–4], and [35, Thm. 1(i)] to our setting in Section 5. All proofs were human-verified, reformulated to improve clarity, and hand-typed. Additionally, Claude

constructed the codes of Examples 19, 28 and 30 in SageMath [29, 37] and verified the quasi-cyclic and MDS properties.